Solar Energy 315 (2026) 114832

Contents lists available at ScienceDirect

Solar Energy

journal homepage: www.elsevier.com/locate/solener

Impact of numerical precision on parameter sensitivity in solar cell models

Martin Ćalasan *, Snežana Vujošević

Faculty of Electrical Engineering, University of Montenegro, Podgorica, Montenegro

ARTICLE INFO

Keywords:
Photovoltaic modeling
Single-diode model
Parameter sensitivity
Numerical precision
Parameter truncation

ABSTRACT

Accurate solar cell and photovoltaic module modeling relies on numerical parameter extraction, yet the influence of parameter precision is rarely examined. This paper provides a critical assessment of how numerical truncation and limited significant digits affect the modeled behaviour of solar cells and PV modules. First, the effect of individual parameter truncation on the root-mean-square error between the experimental and modeled characteristics is quantified, and a normalized sensitivity coefficient is introduced to compare parameter effects on a common dimensionless scale. Second, pairs of parameters are simultaneously reduced to evaluate whether numerical interactions amplify or compensate for modeling errors. In addition, controlled percentage parameter perturbations are analyzed as a complementary sensitivity test, while a Monte Carlo current-noise analysis is performed to assess whether truncation-induced RMSE changes are practically meaningful relative to estimated measurement uncertainty. The results reveal a clear imbalance: the ideality factor, photocurrent, and, in some cases, saturation current and series resistance require higher numerical precision, while parallel resistance remains largely insensitive and contributes negligibly to accuracy degradation. Furthermore, fixing the parallel resistance to different precision levels and re-estimating the remaining parameters produces almost identical model errors, confirming its limited role in numerical fidelity. These findings demonstrate that commonly used high-precision parameter sets may contain redundant digits for certain parameters, whereas insufficient precision for others can distort model accuracy and physical interpretation. Therefore, the proposed precision thresholds should be interpreted as practical guidelines for the analyzed benchmark cases rather than as universal fixed limits.

1. Introduction

The electric power sector is undergoing a major transition driven by climate change, fossil fuel limitations, and the need for sustainable development [1]. In this context, renewable energy sources are increasingly recognized as essential for achieving energy security and reducing environmental impact [2]. Among them, solar energy has a particularly important role because of its wide availability, high technical potential, and applicability across scales ranging from distributed systems to large photovoltaic power plants [3,4]. Rapid progress in photovoltaic (PV) technologies has reduced costs, improved efficiency, and accelerated grid integration, making solar energy one of the key contributors to the ongoing energy transition [5,6].

The growing deployment of PV systems, especially in decentralized applications, increases the need for accurate modelling of solar cells and modules under different operating conditions. Such modelling is important not only for simulation and diagnostics, but also for the planning and operation of modern power systems with a high share of distributed generation [3,7]. For this reason, improving the numerical reliability of PV model parameterization remains an important research objective.

A wide range of models for describing the current–voltage (I–V) characteristics of PV cells and modules has been developed in the literature. To this end, various approaches are employed, from simple empirical equivalent circuits, through the single-diode models (SDM) [8] and two-diode models (DDM) [9], up to multi-diode schemes with three and four diodes (TDM [9], FDM [10,11]), as well as detailed physical models based on solving the carrier transport equations in the semiconductor. The introduction of additional diodes in TDM/FDM models enables a finer separation of different recombination mechanisms and often results in better fitting under extreme operating conditions (very low irradiance, pronounced degradation, hot spots, etc.), but at the cost of a significantly larger number of parameters, stronger nonlinearity, and a more complex identification procedure. Moreover,

* Corresponding author.
E-mail address: martinc@ucg.ac.me (M. Ćalasan).

https://doi.org/10.1016/j.solener.2026.114832
Received 6 December 2025; Received in revised form 6 June 2026; Accepted 24 June 2026
Available online 1 July 2026

modifications of the standard equivalent-circuit structures of solar cells can contribute to a more accurate representation of their current–voltage characteristics and, consequently, to improved solar cell models [12–15].

In the context of engineering calculations, where the goal is to describe the behaviour of photovoltaic modules under standard operating conditions in a reliable yet numerically efficient manner, the five-parameter single-diode model (SDM) emerges as the most suitable compromise between physical soundness and simplicity [16,17]. The model employs five effective parameters: the photogenerated current I_{PV}, the diode reverse saturation current I_0, the diode ideality factor n, the series resistance R_S, and the shunt resistance R_P. These parameters concisely capture the dominant loss mechanisms in the *p–n* junction, series contacts, and conductive paths within the cell, while at the same time enabling consistent scaling of the model with changes in irradiance and temperature. This combination of moderate complexity (compared to TDM/FDM) and the practical need for numerical parameter identification makes the SDM a convenient reference framework for analysing the sensitivity, robustness, and accuracy of the parameters, which is the main focus of this work.

However, a significant limitation in the application of this model, as well as other equivalent-circuit models, is the fact that its parameters are not directly available from the datasheet. Manufacturers typically provide only a few characteristic operating points of the module—short-circuit current, open-circuit voltage, and the maximum power point—while the model parameters remain "hidden" [18,19]. The parameters therefore have to be estimated numerically by determining the set of values for which the single-diode model most accurately reproduces the measured I–V characteristic under given irradiance and temperature conditions.

The standard approach for solar cell parameter estimation is based on defining an error function, most commonly the root-mean-square error (RMSE) between the measured currents and the currents obtained from the model for the corresponding set of voltages [20]. The estimation problem is then reduced to the numerical minimization of this function. In the context of the SDM, this may at first glance appear to be a relatively simple problem with five unknowns; however, the nonlinear structure of the SDM equation, the presence of the exponential term, and potential correlations among the parameters make this optimization numerically sensitive. Precisely because of the nonlinearity of the equation and the presence of multiple parameters, this problem has become a "benchmark test" for a large number of optimization algorithms.

During the optimization process, such algorithms iteratively update the parameter values according to predefined search rules in order to minimize the selected error function. However, in practical numerical environments the representation and storage of numerical values are limited by predefined precision levels, typically expressed through the number of decimal digits. For example, programming environments such as MATLAB allow different formats for numerical representation (e.g., format short and format long), which determine the number of displayed digits. Therefore, the objective of this work is not to analyze the convergence properties of optimization algorithms themselves, but rather to investigate how the numerical precision of the obtained parameters influences the accuracy of the photovoltaic model.

Against this background, several recent contributions have sought to systematize the rapidly growing literature on PV parameter estimation and to provide a clearer picture of recent developments in this area. In [21], a comprehensive overview of parameter estimation methods for the SDM is given, covering both classical numerical procedures and modern optimization-based approaches, in particular metaheuristic methods, with a comparison of accuracy and computational cost burden. Reference [22] focuses on explicit models for describing the I–V characteristics of PV cells, while [23] reviews current trends and future research directions in parameter extraction for SDM, DDM, and TDM models. Building on these review papers, numerous authors have proposed new or improved metaheuristic algorithms for parameter identification, including, among others, the atomic orbital search and an enhanced JAYA algorithm [24,25]. A particularly important contribution is [26], where a larger number of recent metaheuristic algorithms are systematically compared on a consistently defined set of test cases, thereby further establishing the PV parameter estimation problem as a reference benchmark for this class of methods. A separate group of studies [27–29] focuses on parameter estimation for the triple-diode model using a butterfly algorithm with a chaos-learning strategy, the Northern Goshawk optimization, and an improved artificial gorilla troops algorithm. In addition, [30–32] consider models of solar cells and modules with multiple diodes and hybrid metaheuristic schemes, providing detailed comparative studies of several methods, which generally report slightly reduced RMSE and/or faster convergence compared with earlier approaches. In [33], an in-depth analysis of parameter estimation for the single-diode PV module model is presented, including a detailed examination of the physical meaning of each parameter, its influence on the I–V characteristics, and a quantitative sensitivity study with respect to the open-circuit voltage and short-circuit current. Based on different algorithms applied to a public PV module dataset, it is shown that several distinct parameter combinations can yield almost identical I–V curves, indicating that the estimated parameters are generally non-unique and may not directly correspond to the true physical parameter values. Finally, [34] addresses parameter estimation for the Bishop model of a photovoltaic cell, suitable for representing both forward and reverse operating regimes, and thus offering a more flexible framework for modelling different operating conditions. Overall, the available literature confirms that PV model parameter estimation has become a standard testbed for metaheuristic algorithms, while also indicating that many open questions remain regarding model structure, parameter sensitivity, and the actual need for high numerical precision.

However, despite the large number of proposed algorithms, the numerical aspect of the problem itself is rarely subjected to critical analysis. There are almost no studies that systematically examine what level of parameter truncation is actually required, how the choice of physical constants and thermal voltage affects the obtained results, or which parameters are truly the most sensitive and which can be represented with reduced numerical precision without a noticeable loss of model accuracy.

In fact, the majority of studies on numerical extraction of PV model parameters exhibit similar methodological assumptions and limitations:

- an exact (often rounded) value of the thermal voltage V_t and the number of series-connected cells is assumed,
- the physically and numerically justified number of significant digits of the model parameters, and, moreover, how many of these digits are actually measurable with real instruments, is not examined, and
- parameter identifiability is rarely discussed.

While substantial attention has been devoted to the optimization mechanism, far less has been devoted to the model itself and its robustness. Consequently, the structure of the underlying numerical problem has remained insufficiently analyzed. This paper is specifically aimed at addressing this neglected aspect of previous research.

Building on these observations, the aim of this paper is to quantitatively investigate the numerical robustness of the single-diode model and to provide clear answers to the following key questions:

How many significant digits are actually required for the SDM parameters in order to faithfully reproduce the I–V characteristic?
Which parameters have a dominant, and which have a negligible influence on the current–voltage characteristics?

For this purpose, the paper focuses on decimal-place analysis, that is, on examining the influence of numerical truncation of the model parameters, as well as on the influence of percentage reduction of each parameter value on the current–voltage characteristics of solar cells/modules. For each such modified parameter set, the I–V characteristic is recalculated and the RMSE is determined with respect to the reference measurement data. In this way, a direct relationship is obtained between the number of significant digits, or the reduction in parameter values, and the deterioration of the agreement between the model and the experimental data, both for each parameter individually and for their combinations.

The main scientific contributions of this paper are as follows:

1. The paper quantitatively analyzes the influence of decimal-place truncation of individual SDM parameters on model accuracy and introduces a normalized sensitivity coefficient to express the relative RMSE change per relative parameter variation.
2. Furthermore, the paper examines the simultaneous reduction of two parameter values, revealing cumulative and compensating interaction effects and providing deeper insight into nonlinear parameter interdependence within the SDM.
3. The practical significance of parameter-reduction-induced RMSE changes is evaluated relative to estimated current-measurement uncertainty through a Monte Carlo current-noise analysis.
4. The paper identifies the key parameters for which numerical precision significantly affects model accuracy, as well as those for which the effect is negligible.

The structure of the paper is as follows: Section 2 briefly reviews the single-diode model (SDM) of a solar cell and its analytical formulation, including the use of the Lambert W function and the RMSE metric for model evaluation. Section 3 introduces the concept of numerical truncation and presents a detailed analysis of the sensitivity of individual SDM parameters on the RTC France solar cell and the MSX-60, Photowatt-PWP 201, CdTe75669, and KC200GT photovoltaic modules. Section 4 examines the simultaneous reduction of two parameter values, revealing cumulative and compensating interaction effects and providing deeper insight into nonlinear parameter interdependence within the SDM. Section 5 focuses on fixing the parallel resistance and addresses the question of whether high numerical precision of parallel resistance is really needed. Section 6 provides a discussion of the results and outlines the main limitations of the paper. Finally, Section 7 discusses the main findings, highlights their practical implications for PV modelling, and summarizes the key conclusions.

2. Mathematical model of the solar cell

Several different models have been proposed in the literature for describing the I–V characteristics of PV cells and modules, ranging from simple equivalent circuits to advanced multi-diode and detailed physical models. Although these models offer greater flexibility and allow a more refined representation of specific effects, they simultaneously introduce a larger number of parameters and additional complexity in their identification.

In this paper, the focus is deliberately confined to the classical five-parameter single-diode model (SDM), which represents the basic and most commonly used reference model in PV system analysis. All developments introduced in the following sections—analysis of significant digits, investigation of numerical robustness, and sensitivity analysis—are considered within the framework of the SDM, as the fundamental model on which many improved approaches are based.

The classic SDM model consists of a direct current source (I_{PV}), a diode (D) connected in parallel with it, and two resistors (R_P and R_S) in parallel and series configurations, respectively.

The equivalent electrical circuit of the single-diode solar cell model is shown in Fig. 1. Mathematically, the relationship between current and voltage in this model can be described by the following equation:

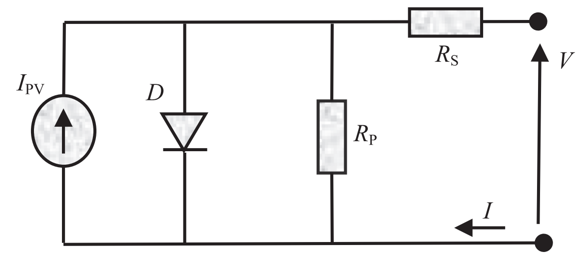


Fig. 1. The single-diode solar cell model.

$I = I_{PV} - I_0\left(e^{\frac{V+IR_S}{nV_{th}}} - 1\right) - \frac{V+IR_S}{R_P}$, (1) where:

I_{PV} – the photogenerated current,

I_0 – the reverse saturation current of the diode,

n – the diode ideality factor, and

$V_{th} = K_B \cdot T/q$ – the thermal voltage (K_B is the Boltzmann constant, T is the temperature in Kelvins and q is the electron charge). For the RTC France solar cell, $N_S = 1$, so V_{th} corresponds directly to the cell thermal voltage. For PV modules, the effect of the number of series-connected cells N_S is included by interpreting V_{th} in Eqs. (1)–(3) as the effective thermal voltage of the considered module, i.e., $V_{th} = N_S \cdot K_B \cdot T/q$. In this formulation, n remains the diode ideality factor of an individual cell, while R_S and R_P denote the equivalent series and shunt resistances of the considered cell/module. Accordingly, Eqs. (1)–(3) are used in a cell-level form for the RTC France solar cell and in an equivalent module-level form for PV modules.

In accordance with the standard single-diode model, the diode saturation current of solar cells is typically in the microampere range, the shunt resistance is significantly larger than the series resistance, and the diode ideality factor n usually takes values between 1 and 2.

Although the basic SDM equation is implicit and nonlinear with respect to the current, it is well known that its exact solution can be obtained in analytical form by using the Lambert W function. This makes it possible to express the output current I directly as a function of the voltage V, without the need for iterative procedures.

The analytical expression is given by the following relation:

$$I = \frac{R_P(I_{PV} + I_0) - V}{R_S + R_P} - \frac{nV_{th}}{R_S} W(x) \quad (2)$$

where:

$$x = \frac{R_S R_P I_0}{nV_{th}(R_S + R_P)} \exp\left(\frac{R_P[V + R_S(I_{PV} + I_0)]}{nV_{th}(R_S + R_P)}\right) \quad (3)$$

and W represents the solution to the Lambert W function.

Based on the analytical solution given by (2), it is possible to directly compute the current for each voltage value in the set of experimental data. This formulation enables accurate tracking of the cell behaviour along the I–V curve and provides a basis for analysing its response under different operating regimes, which is of particular importance for performance optimization and model evaluation.

To quantitatively assess the accuracy of the model with respect to the experimental data, the root-mean-square error (RMSE) is used. This metric represents a standard criterion for evaluating the deviation between measured and modelled current values and is defined by the following expression:

$$RMSE = \sqrt{\frac{1}{N_m}\sum_{i=1}^{N_m}\left(I_i^{meas} - I_i^{calc}\right)^2} \quad (4)$$

where N_m represents the number of measured points, I^{meas} is the measured current value for a certain voltage, V^{meas} is the measured voltage value.

2.1. Parameter precision and the importance of numerical truncation

In this paper, several key aspects of solar cell modelling and parameter estimation are analysed, with particular emphasis on the effect of numerical truncation in specifying parameter values. Truncation denotes the process of reducing the number of significant digits in numerical values, retaining only those digits considered relevant while discarding the rest. This practice is common in numerical modelling in order to reduce computational complexity and control data precision. However, limiting the number of significant digits can introduce errors, since the original parameters are approximated with reduced precision, which can directly affect the accuracy of the model.

Parameters of solar cells, such as the photocurrent, diode reverse saturation current, series resistance, shunt resistance, and ideality factor, are most commonly derived from experimental I–V characteristics using numerical optimization methods, such as the least-squares method, whereby these values are obtained with a large number of decimal places in order to increase numerical precision. However, high numerical precision does not necessarily guarantee high model accuracy, which can be attributed to several factors:

- the models are often simplified and cannot fully capture all real physical processes;
- experimental measurements inherently contain errors and noise;
- excessive numerical precision may lead to numerical issues in the optimization process; and
- some parameters have only a limited influence on the output characteristics of the model, so additional precision in their values has no significant effect on the results.

Therefore, it is of crucial importance to determine the optimal number of significant digits for each parameter in order to achieve an appropriate balance between model accuracy and numerical stability, while avoiding unnecessary increases in computational demands. In this context, the analysis of how the numerical precision of the parameters affects the behaviour of the model represents one of the central contributions of this paper.

3. Effect of individual parameter truncation on modeling accuracy

For the purpose of quantitatively analysing the impact of the number of significant digits on the accuracy of the single-diode model, the analysis was carried out on five representative photovoltaic structures: the RTC France solar cell, the Solarex MSX-60 PV module, the Photowatt-PWP 201 PV module, the CdTe75669 thin-film PV module, and the KC200GT PV module. These samples were selected due to their technological diversity, their widespread use in the literature, and the availability of reference measured I–V data, which enables a comprehensive assessment of the effects of numerical truncation across different types of PV systems, from individual cells to commercial modules.

In addition, in order to provide a more detailed assessment of the relative contribution of individual parameters, an additional normalized interpretation of the decimal-place truncation results for all SDM parameters was introduced. This approach enables parameter changes of different orders of magnitude to be compared on a common dimensionless scale.

The reference value of the considered parameter is denoted by p_0, while p_{trunc} represents the value of the same parameter after decimal-place truncation. Analogously, RMSE_0 denotes the error value obtained using the reference parameter set, whereas RMSEtrunc represents the error value after decimal-place truncation of the considered parameter.

The relative change in the parameter value is defined as:

$$\frac{\Delta p}{p_0} = \frac{p_{trunc} - p_0}{p_0} \tag{5}$$

whereas the relative change in the error is determined by the following relation:

$$\frac{\Delta RMSE}{RMSE_0} = \frac{RMSE_{trunc} - RMSE_0}{RMSE_0} \tag{6}$$

Using these quantities, the normalized sensitivity coefficient S_{p} is defined, indicating the relative change in RMSE with respect to the relative change in the considered parameter:

$$S_p = \frac{\Delta RMSE/RMSE_0}{\Delta p/p_0} \tag{7}$$

In this way, the obtained coefficient S_{p} provides a measure of the model sensitivity to variations in an individual parameter, independently of the parameter's absolute value.

3.1. RTC France solar cell

The RTC France cell is a monocrystalline silicon solar cell that is frequently used as a reference sample in academic and experimental analyses of photovoltaic models. This cell is characterized by a well-documented I–V characteristic, stable performance, and extensive coverage in the literature, which makes it suitable for the evaluation of numerical models and the validation of optimization algorithms.

The basic reference points used in the analysis are given in Table 1. Detailed measured I–V data for the RTC France cell are presented in Appendix A (Table A1).

Table 2 presents the parameter values of the RTC France solar cell collected from a number of relevant literature sources [8,12,35–37]. In this study, the parameters obtained using the Walrus Optimization Algorithm (WaOA) and Bacterial Foraging Optimization (BFO) algorithms were used for the subsequent analysis. It can be observed that the selected parameter values lie within the range of values reported in the literature for this solar cell, confirming that the adopted parameter sets represent a suitable and representative basis for the analyses carried out in this work. The resulting parameter sets are subsequently used to analyse the impact of parameter truncation in terms of significant digits, which will be discussed in detail in the following sections.

In order to investigate the impact of numerical truncation on model accuracy, the number of significant digits of individual parameters was gradually reduced (from eight down to three). In each case, the remaining parameters were kept at full precision, and the modified model was used to recompute the I–V characteristic. The resulting output current values were compared with the reference measured data, and the root-mean-square error (RMSE) was used as a measure of the deviation. This approach enables a quantitative assessment of the sensitivity of the model to a reduced number of decimal places for each parameter.

Fig. 2 illustrates the impact of reducing the number of decimal places of individual parameters (each considered separately) on the model accuracy, expressed in terms of the RMSE values, for the parameter set obtained using the Walrus Optimization Algorithm. Fig. 2a shows the absolute RMSE as a function of the number of decimal places (from 8

Table 1
Reference electrical parameters for the RTC France cell under STC conditions.

Parameter	Notation	Value
Short-circuit current	I_{SC}	0.7603
Open-circuit voltage	V_{OC}	0.5728
Current at maximum power point	I_{MPP}	0.6894
Voltage at maximum power point	V_{MPP}	0.4507
Maximum output power	P_{MPP}	0.3107

Table 2
Literature-reported SDM parameter values for the RTC France solar cell.

Ref.	Method	I_{PV} [A]	I_0 [μA]	n	R_S [Ω]	R_P [Ω]
[8]	WaOA*	0.76078796	0.31068404	1.47726761	0.03654695	52.88969619
[8]	BFO*	0.76079554	0.31062347	1.47724953	0.03654677	52.80489646
[12]	C-POA	0.76080423	0.30475032	1.47533967	0.03663756	52.46402289
[35]	BA-CR	0.760788	0.310691	1.47727	0.036547	52.8899
[36]	HOA	0.760746	0.347	1.48840	0.0361	55.5
[36]	EHO	0.76063	0.361	1.49260	0.0358	56.9
[37]	SOA	0.760775531	0.32302091	1.48118362	0.03637709	53.7185274
Literature range		0.76063–0.76080423	0.30475032–0.361	1.47533967–1.49260	0.0358–0.03663756	52.46402289–56.9
Comment on the parameter values used in this study		within reported range	within reported range	within reported range	within reported range	within reported range

* The values marked with an asterisk represent the reference optimized SDM parameter sets used in the subsequent decimal-place truncation, sensitivity, and pairwise interaction analyses.

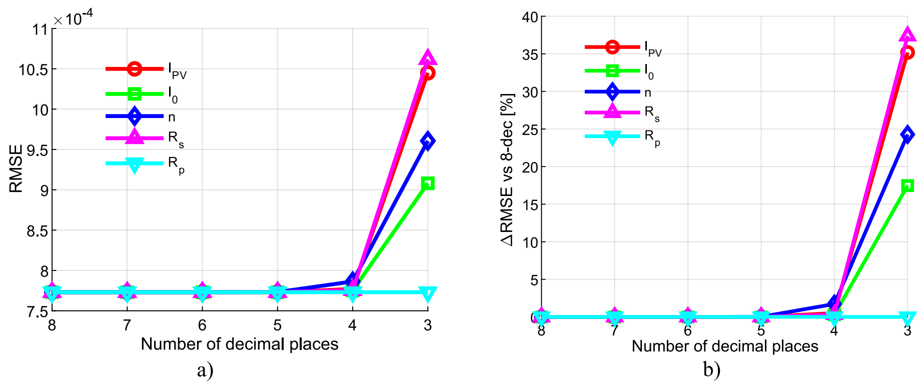


Fig. 2. Impact of parameter truncation on SDM accuracy for the RTC France cell. a) RMSE vs. number of decimals, b) relative RMSE increase (WaOA).

down to 3), while Fig. 2b presents the corresponding relative change in RMSE with respect to the reference case with 8 decimal places. The corresponding numerical results are presented in Appendix A (Table A2).

To further assess the robustness of the model, Fig. 3 shows the effect of truncating individual parameters when the initial values are obtained using the BFO optimization algorithm. Fig. 3a presents the variation of the absolute RMSE as the number of decimal places is reduced from 8 to 3, while Fig. 3b shows the corresponding relative change in error with respect to the reference, highest-precision case. This representation provides insight into how the model behaves under the same numerical constraints when the parameters are estimated by a different optimization approach.

In both analysed cases, the results show that the parameters R_S, I_{PV}, n and I_0 are particularly sensitive to a reduction in the number of decimal places — already at three decimal places the error increases sharply. By contrast, R_P exhibits pronounced numerical stability, since the RMSE remains almost unchanged even when it is rounded to only three decimal places.

These findings clearly indicate the need for differentiated precision in numerical modelling: certain parameters require high accuracy in order to preserve the fidelity of the model, whereas others can be simplified without a significant impact on the model performance.

In addition to the previous analysis, an additional study of numerical truncation from the perspective of significant digits was carried out. For this purpose, the RTC cell parameters obtained using the WaOA algorithm were employed. The corresponding results are presented in Fig. 4, while the numerical values are given in Appendix A (Table A3). It can be observed that the results obtained on the basis of significant digits are qualitatively very similar to those obtained by simple truncation of decimal places. In particular, the same overall ranking of parameter sensitivity and the same trend in RMSE variation are preserved, indicating that the main conclusions of the study remain robust under this alternative approach to precision reduction.

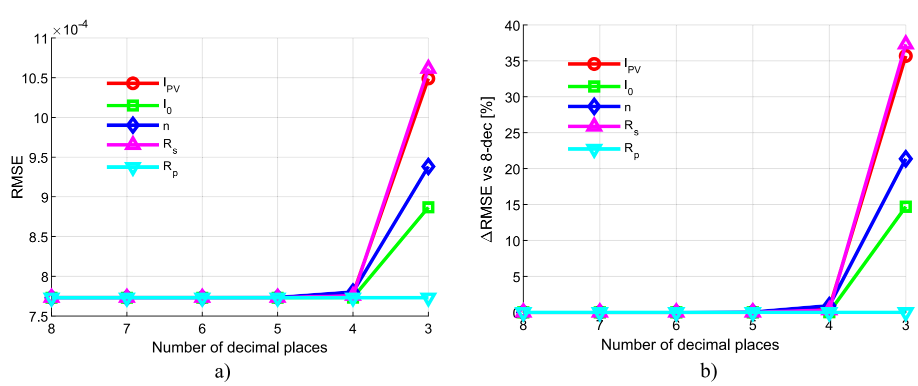
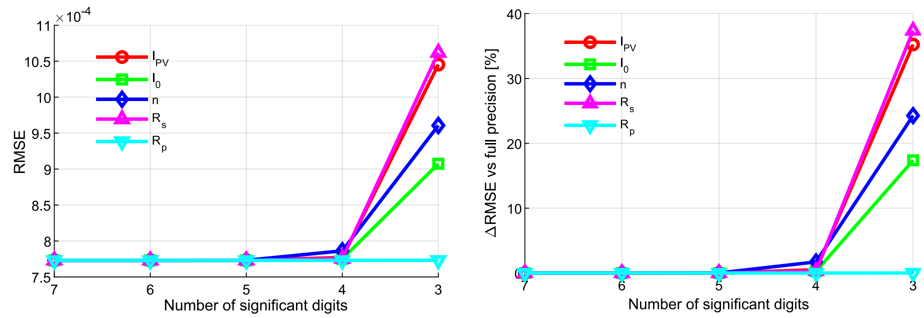


Fig. 3. Impact of parameter truncation on SDM accuracy for the RTC France cell a) RMSE vs. number of decimals, b) relative RMSE increase (BFO).



a) b)

Fig. 4. Impact of parameter significant-digit precision on SDM accuracy for the RTC France cell a) RMSE vs. number of significant digits, b) relative RMSE increase with respect to full precision (WaOA).

In addition to RMSE, other performance metrics, namely the mean absolute error (MAE), mean absolute percentage error (MAPE), and coefficient of determination (R^2), were also calculated for the RTC France solar cell using the parameter set obtained by the WaOA algorithm. All corresponding results are presented in Table 3.

The results in Table 3 show that all considered metrics are mutually consistent and indicate the same general trend in model accuracy as numerical truncation is applied. The RMSE and MAE values show that the model is practically insensitive to parameter truncation down to about 6–5 decimal places for most parameters, whereas a more pronounced deterioration appears at 4 and especially at 3 decimal places. In this respect, RMSE increases more rapidly than MAE, indicating that parameter truncation does not affect all points of the I–V characteristic equally, but rather amplifies certain larger deviations.

MAPE confirms the same overall trend, but also shows that the relative error does not necessarily increase with the same intensity as RMSE and MAE, since it depends on the ratio between the error and the measured current value. Therefore, for some parameters it can be observed that RMSE increases while MAPE remains more stable or even slightly decreases, which means that absolute and relative deviations do not always follow exactly the same behavior. On the other hand, the coefficient of determination R^2 remains very close to unity in all cases, indicating that the model preserves an excellent overall agreement with the measured data even under parameter truncation. Nevertheless, small changes in R^2 do not imply that the effect of parameter truncation is negligible, because RMSE, MAE, and MAPE reveal more sensitive changes in the local accuracy of the approximation.

3.2. *MSX-60 solar module*

As a second representative example, the impact of parameter truncation is analysed for the parameters of the MSX-60 solar module, which is frequently used as a reference in the literature due to the availability of experimental data and its widespread use in studies of photovoltaic models.

In contrast to the previously analysed RTC France cell, which represents a single device, the MSX-60 is a commercial photovoltaic module composed of multiple series-connected cells, allowing the numerical robustness of the model to be tested under more realistic operating conditions.

The basic reference points used in the analysis are given in Table 4.

Table 4
Reference electrical parameters for the MSX-60 solar module under STC conditions.

Parameter	Notation	Value
Short-circuit current	I_{SC}	3.80
Open-circuit voltage	V_{OC}	21.1
Current at the maximum power point	I_{MPP}	3.50
Voltage at the maximum power point	V_{MPP}	17.1
Maximum output power	P_{MPP}	59.85

Table 3
RMSE, MAE, MAPE, and R^2 values for the RTC France solar cell using the WaOA parameter set under decimal-place truncation (The values for RMSE and MAE have been multiplied by 10,000).

	Parameter	Decimal places					
		8	7	6	5	4	3
RMSE	I_{PV}	7.7300626905	7.7300627139	7.7300675148	7.7303896856	7.7698189484	10.4497981821
	I_0	7.7300626905	7.7300626958	7.7300626958	7.7301138792	7.7521752817	9.0808222069
	n	7.7300626905	7.7300626932	7.7300735978	7.7317618353	7.8630964480	9.6064228396
	R_S	7.7300626905	7.7300627186	7.7300729761	7.7306135923	7.7551749999	10.6193759006
	R_P	7.7300626905	7.7300626905	7.7300626905	7.7300626905	7.7300626920	7.7300627397
MAE	I_{PV}	6.7818317311	6.7818049885	6.7814038497	6.7782838780	6.8071767016	8.2869706649
	I_0	6.7818317311	6.7818761068	6.7818761068	6.7863136966	6.8750726802	7.6033832840
	n	6.7818317311	6.7817978135	6.7797627590	6.7635023020	6.7204419076	7.5283922157
	R_S	6.7818317311	6.7818876202	6.7828936322	6.7896007751	6.8343326512	8.6127795144
	R_P	6.7818317311	6.7818317290	6.7818317268	6.7818315927	6.7818295817	6.7818161744
MAPE	I_{PV}	0.4424414642	0.4424284679	0.4422335237	0.4407172899	0.4253308198	0.3110909252
	I_0	0.4424414642	0.4424623299	0.4424623299	0.4445489055	0.4862841043	0.8019483724
	n	0.4424414642	0.4424253451	0.4414581975	0.4305275646	0.3413674080	0.4366560485
	R_S	0.4424414642	0.4424414851	0.4424418604	0.4424443512	0.4424604493	0.4764507080
	R_P	0.4424414642	0.4424414603	0.4424414559	0.4424411953	0.4424372862	0.44241122493
R^2	I_{PV}	0.9999934274	0.9999934274	0.9999934274	0.9999934268	0.9999933596	0.9999879888
	I_0	0.9999934274	0.9999934274	0.9999934274	0.9999934273	0.9999933897	0.9999909297
	n	0.9999934274	0.9999934274	0.9999934274	0.9999934245	0.9999931992	0.9999898494
	R_S	0.9999934274	0.9999934274	0.9999934274	0.9999934265	0.9999933846	0.9999875958
	R_P	0.9999934274	0.9999934274	0.9999934274	0.9999934274	0.9999934274	0.9999934274

Detailed measured I–V data for the MSX-60 solar module are presented in Appendix B (Table B1).

Table 5 presents the parameter values of the Solarex MSX-60 PV module reported in the literature and obtained using different optimization approaches. In this study, the parameter set obtained using the Adaptive Walrus Optimization Algorithm (AWaOA) was used for the subsequent numerical precision and sensitivity analyses. It can be observed that the selected parameter values lie within the range of values reported in the literature for this module, confirming that the adopted parameter set represents a suitable and representative basis for the analyses carried out in this work.

Fig. 5 illustrates the impact of reducing the number of decimal places of individual parameters for the MSX-60 solar module on the model accuracy, expressed in terms of the RMSE values, for the parameter set obtained using the Adaptive Walrus Optimization Algorithm (AWaOA). Fig. 5a shows the absolute RMSE as a function of the number of decimal places (from 8 down to 4), while Fig. 5b presents the corresponding relative change in RMSE with respect to the reference case with 8 decimal places. The corresponding numerical results are presented in Appendix B (Table B2).

The results obtained for the MSX-60 solar module further confirm that the model parameters exhibit markedly different sensitivity to numerical truncation. In this case, the parameters n and I_0 are highly sensitive to numerical truncation—already when the precision is reduced to four decimal places, a noticeable increase in RMSE is observed, indicating the need for high precision in their specification.

These results further highlight the importance of a differentiated approach to numerical modelling: while some parameters require precise identification, others can be approximated with a smaller number of significant digits without major consequences, thereby reducing computational requirements without compromising the accuracy of the model.

In addition to the previous analysis, an additional study of parameter truncation from the perspective of significant digits was also carried out for the MSX-60 solar module. For this purpose, the parameter set obtained using the AWaOA algorithm was employed. The corresponding results are presented in Appendix B, where Table B3 gives the results under significant-digit truncation. It can be observed that the analysis based on significant digits leads to conclusions that are qualitatively consistent with those obtained by simple decimal-place truncation. In particular, the same general pattern of parameter sensitivity is preserved, with n and I_0 showing the highest sensitivity, while R_P remains the least sensitive parameter. Although the significant-digit-based reduction produces somewhat stronger RMSE growth at lower precision levels, especially at 4 and 3 significant digits, the overall sensitivity ranking and the main conclusions of the study remain unchanged, confirming the robustness of the results with respect to the adopted precision-reduction criterion.

3.3. Photowatt-PWP 201 solar module

As the third case in the analysis, model parameters for the Photowatt-PWP 201 solar module were used. This module is also a commercially available solution based on polycrystalline silicon technology. Including this module in the analysis extends the evaluation to another realistic photovoltaic system, thereby providing an additional verification of the consistency of the results under different experimental conditions and modelling setups.

The key reference points used in the analysis are presented in Table 6. Detailed measured I–V data for the Photowatt-PWP 201 solar module are given in Appendix C (Table C1). Table 7 presents the parameter values of the Photowatt-PWP 201 PV module reported in the literature and obtained using different optimization approaches. In this study, the parameter set obtained using the Chaotic Gorilla Troops Optimization (CGTO) algorithm, taken from [41], was used for the subsequent numerical precision and sensitivity analyses. The selected parameter values are within the range of values reported in the literature for this module, confirming that the adopted parameter set provides a suitable and representative reference basis for the analyses carried out in this work. Therefore, Table 7 is used to further analyse the sensitivity of the model to a reduced number of significant digits, while retaining the same methodological approach as in the previous analyses.

Fig. 6 shows the impact of reducing the number of decimal places of individual parameters on the model accuracy for the Photowatt-PWP 201 solar module. The analysis is based on parameters obtained using the Chaotic Gorilla Troops Optimization (CGTO) algorithm, and the modelling error is quantified by the RMSE metric. Fig. 6a depicts the variation of the absolute RMSE as a function of the number of decimal places (from 8 down to 3), while Fig. 6b illustrates the corresponding relative change with respect to the reference value at full precision. The corresponding numerical results are presented in Appendix C (Table C2).

The results shown in Fig. 6 indicate that the accuracy of the model for the Photowatt-PWP 201 solar module is particularly sensitive to the parameter truncation of the diode ideality factor n. Although the RMSE remains almost unchanged down to four decimal places, reducing the precision to three decimal places leads to a sharp increase in the error, which points to a critical precision threshold for this parameter. The other parameters – the photogenerated current I_{PV}, saturation current I_0, series resistance R_S, and especially the shunt resistance R_P – exhibit much higher numerical stability, with the RMSE remaining nearly unchanged even under stronger rounding. This analysis further confirms that the parameters do not require uniform numerical precision; instead, the required precision depends on their sensitivity to the overall accuracy of the model.

In addition to the previous analysis, an additional study of parameter truncation from the perspective of significant digits was also carried out for the Photowatt-PWP 201 solar module. For this purpose, the parameter set obtained using the CGTO algorithm was employed for decimal-place truncation, while the significant-digit-based analysis is presented

Table 5
Literature-reported SDM parameter values for the MSX-60 solar module.

Ref.	Method	I_{PV} [A]	I_0 [μA]	n	R_S [Ω]	R_P [Ω]
[8]	AWaOA*	3.81097398	0.12057351	1.32109677	0.22786515	894.28671832
[38]	GTO-HBA	3.8127	0.14051	1.332513	0.22351	1105.5869
[38]	HBA-GTO	3.81268	0.14	1.3325	0.2235	1155.6258
[7]	NM	3.8084	4.8723e-04	1.0003	0.3692	169.0471
[39]	BC	3.808	1.22e-3	1.045	0.316	146.08
[40]	A&I	3.7983	6.79e-2	1.28	0.251	582.727
Literature range		3.7983–3.8127	4.8723e-04–0.14051	1.0003–1.332513	0.2235–0.3692	146.08–1155.6258
Comment on the parameter values used in this study		within the reported range	within the reported range	within the reported range	within the reported range	within the reported range

* The parameter values marked with an asterisk represent the reference optimized SDM parameter sets used in the subsequent decimal-place truncation, sensitivity, and pairwise interaction analyses.

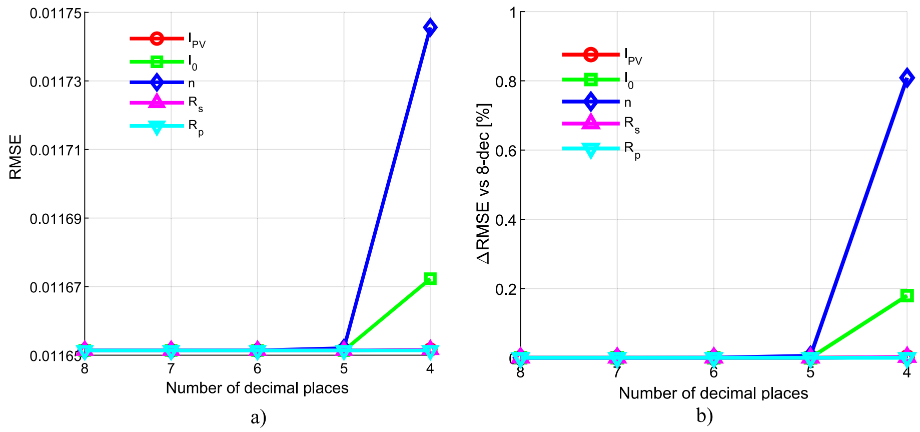


Fig. 5. Impact of parameter truncation on SDM accuracy for the MSX-60 solar module, a) RMSE vs. number of decimals, b) relative RMSE increase (AWaOA).

Table 6
Reference electrical parameters for the Photowatt-PWP 201 solar module under STC conditions.

Parameter	Notation	Value
Short-circuit current	I_{SC}	1.03
Open-circuit voltage	V_{OC}	16.778
Current at the maximum power point	I_{MPP}	0.912
Voltage at the maximum power point	V_{MPP}	12.641
Maximum output power	P_{MPP}	11.536

in Appendix C, where Table C3 summarizes the corresponding RMSE values. It can be observed that the analysis based on significant digits leads to conclusions that are qualitatively consistent with those obtained by simple decimal-place truncation.

3.4. CdTe75669 solar module

As the fourth case in the analysis, model parameters for the CdTe75669 solar module were considered. This module represents a cadmium telluride thin-film photovoltaic device from the Golden benchmark dataset within the NREL PV module performance validation database. In this context, "Golden" refers to the measurement location in Golden, Colorado, rather than to the manufacturer of the module. Including this module in the analysis extends the evaluation beyond crystalline-silicon devices and provides an additional verification of the consistency of the results across different photovoltaic technologies, experimental conditions, and modelling setups. The key reference points

Table 7
Literature-reported SDM parameter values for the Photowatt-PWP 201 solar module.

Ref.	Method	I_{PV} [A]	I_0 [μA]	n	R_S [Ω]	R_P [Ω]
[41]	CGTO*	1.03235759	2.49659566	1.31662653	1.24054732	748.32294187
[42]	ISHTSA	1.03051	3.48226	1.35118972	1.20127	981.98227
[43]	RL-GJO	1.0305	3.48	1.35111111	1.2013	981.98
[44]	OBEDO	1.0305	3.48	1.35118889	1.2013	981.98
[45]	HKOA	1.0324	2.497	1.31660000	1.242	748.3248
[46]	En-PDO	1.0307	2.4052	1.31266667	1.2493	861.29
Literature range		1.0305–1.0324	2.4052–3.48226	1.31266667–1.35118972	1.20127–1.2493	748.3248–981.98227
Comment on the parameter values used in this study		within the reported range	within the reported range	within the reported range	within the reported range	close to reported range

* The parameter values marked with an asterisk represent the reference optimized SDM parameter sets used in the subsequent decimal-place truncation, sensitivity, and pairwise interaction analyses.

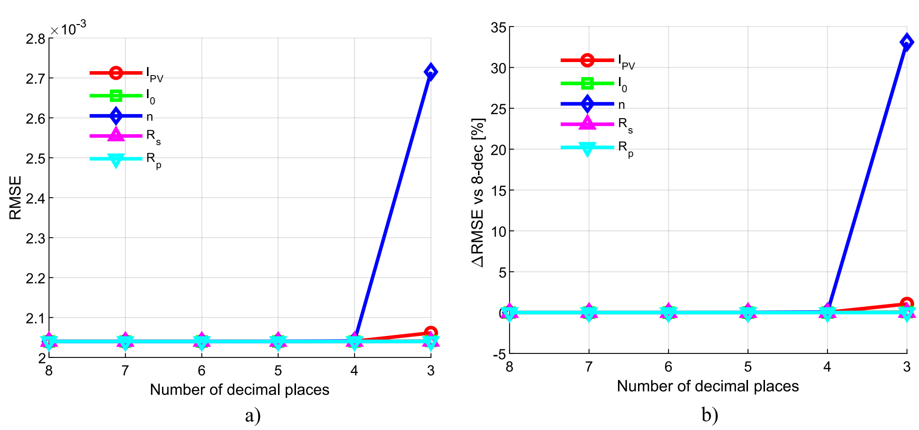


Fig. 6. Impact of parameter truncation on SDM accuracy for the Photowatt-PWP 201 solar module, a) RMSE vs. number of decimals, b) relative RMSE increase (CGTO).

used in the analysis are presented in Table 8. The parameters of the single-diode model were estimated in this study using the ERWCA algorithm, and the resulting parameter set, as well as RMSE, shown in Table 9, is further used to analyse the sensitivity of the model to a reduced numerical precision, while retaining the same methodological approach as in the previous analyses.

Fig. 7 shows the effect of truncating individual parameters on the model accuracy for the CdTe75669 solar module. The analysis is based on the parameter set estimated using the ERWCA algorithm, and the modelling error is quantified by the RMSE metric. In the case of decimal-place truncation, the corresponding absolute and relative RMSE variations are presented in Fig. 7a and Fig. 7b, respectively, while the numerical values are listed in Appendix D (Table D1). In addition, the influence of parameter truncation from the perspective of significant digits was also examined, and the corresponding numerical results are given in Appendix D (Table D2).

The results indicate that the CdTe75669 solar module is particularly sensitive to truncation of the diode ideality factor n, while the photo-generated current I_{PV} also shows noticeable sensitivity when the number of decimal places is strongly reduced. Under decimal-place truncation, the RMSE remains nearly unchanged for most parameters down to 5–4 decimal places, whereas a more pronounced increase appears at 3 decimal places, especially for n. The parameters I_0, R_S, and particularly R_P exhibit much higher numerical stability, with only negligible RMSE variation over the same parameter truncation range. A similar overall pattern is observed when precision is reduced in terms of significant digits: the same ranking of parameter sensitivity is preserved, but the RMSE growth becomes more pronounced at lower precision levels, especially for I_{PV} and n. This confirms that the effect of parameter truncation depends strongly on the physical role of each parameter, and that the required numerical precision is not uniform across all parameters.

3.5. KC200GT solar module

As the fifth case in the analysis, model parameters for the KC200GT solar module were considered. This device is a commercially available polycrystalline photovoltaic module manufactured by Kyocera, with a rated maximum power of 200 W under standard test conditions. The module consists of 54 series-connected cells and is a widely used benchmark in photovoltaic modelling studies because its electrical characteristics are well documented and representative of practical crystalline-silicon PV systems. According to the manufacturer's specifications, its main reference operating points under STC are I_{sc} = 8.21 A, V_{oc} =32.9 V, I_{MP}=7.61 A, V_{MP} =26.3 V, and P_M = 200 W. Including this module in the analysis extends the evaluation to another realistic commercial photovoltaic system and provides an additional verification of the consistency of the results under different technologies, datasets, and modelling conditions.

Table 10 presents the estimated SDM parameter values of the KC200GT solar module obtained using the ERWCA algorithm, together with the parameter values reported in the literature for the same module. It can be observed that the estimated parameters lie within the range of values available in the literature, which confirms the consistency of the obtained parameter set. Therefore, the ERWCA-based parameter set was adopted as a representative reference set for the subsequent decimal-place truncation, sensitivity, and pairwise interaction analyses.

Table 8
Reference electrical parameters for the CdTe75669 solar module under STC conditions.

Parameter	Notation	Value
Short-circuit current	I_{SC}	1.192
Open-circuit voltage	V_{OC}	86.33
Current at maximum power point	I_{MPP}	1.003
Voltage at maximum power point	V_{MPP}	62.17
Maximum output power	P_{MPP}	62.35

Fig. 8 shows the effect of truncating individual parameters on the model accuracy for the KC200GT solar module. The analysis is based on the parameter set estimated using the ERWCA algorithm, and the modelling error is quantified by the RMSE metric. In the case of decimal-place truncation, the corresponding absolute and relative RMSE variations are presented in Fig. 8a and Fig. 8b, respectively, while the numerical values are listed in Appendix E (Table E1). In addition, the influence of parameter truncation from the perspective of significant digits was also examined, and the corresponding numerical results are given in Appendix E (Table E2).

The results for the KC200GT solar module are consistent with those obtained for the previously analyzed cells and modules. The highest sensitivity is again observed for the diode saturation current I_0 and the diode ideality factor n, while I_{PV} and R_S remain more stable, and R_P shows the weakest sensitivity to parameter truncation. Under both decimal-place truncation and significant-digit reduction, the same overall ranking of parameter sensitivity is preserved, with a more noticeable RMSE increase only at lower precision levels. This again confirms that the required numerical precision is parameter-dependent rather than uniform for all SDM parameters.

3.6. Normalized (dimensionless) sensitivity

Table 11 provides a summary overview of the relative change in RMSE for all considered cells and modules with respect to the reference case without digit truncation. In this analysis, the parameter values were truncated from 7 to 4 decimal places for all cells and modules. The results clearly show that the effect of parameter truncation is not the same for all parameters or for all devices. In general, for 7 and 6 decimal places, the relative change in RMSE remains negligibly small, whereas a more pronounced increase in error appears only under stronger parameter truncation, particularly at 4 decimal places. In most cases, the highest sensitivity is observed for the parameters n and I_{PV}, while R_P consistently remains the least sensitive parameter. Overall, the results confirm that the required numerical precision of SDM parameters is not uniform, but depends both on the physical role of the parameter and on the characteristics of the specific solar device.

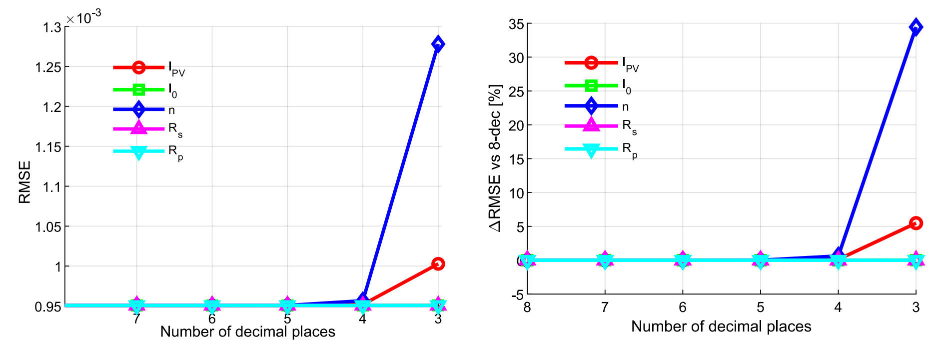
However, in addition to the previously presented analysis, an additional normalized analysis of the influence of decimal-place parameter truncation was also performed. The detailed numerical results of this analysis are provided in the Appendix, in Tables A4, B4, C4, D3, and E3 for the RTC France cell and the MSX-60, Photowatt-PWP 201, CdTe75669, and KC200GT modules, respectively, while the summary results are presented in Table 11. A graphical representation of the obtained results is shown in Fig. 9.

The obtained results confirm the conclusions of the previous analysis, but also enable a clearer comparison of the influence of individual parameters. Based on the values of $|S_p|$, the diode ideality factor n stands out as the most sensitive parameter for all analyzed cells and modules. The highest $|S_p|$ values were obtained precisely for this parameter, indicating that even a relatively small change in its value can cause a pronounced change in the model error. Therefore, when reporting and using the parameter n, a higher number of decimal places should be retained.

The analysis of the relative error change $\Delta RMSE/RMSE_0$ shows that the largest RMSE increase most often occurs when truncating the parameter n, especially for the MSX-60, Photowatt-PWP 201, and CdTe75669 modules. For the RTC France cell, in addition to n, the parameters R_S and I_{PV} also have a very pronounced influence, with R_S and I_{PV} producing the largest RMSE increase under coarse truncation to three decimal places. For the KC200GT module, the largest RMSE increase under truncation to three decimal places occurs for the diode saturation current I_0, while the parameter n still has the highest $|S_p|$ value. This difference shows that $\Delta RMSE/RMSE_0$ and $|S_p|$ provide complementary

Table 9
The estimated values of parameters for CdTe75669 solar module.

Algorithm	I_{PV} [A]	I_0 [μA]	n	R_S[Ω]	R_P[Ω]	RMSE
ERWCA	1.16235192	6.20055140	1.94879954	7.02217388	1624.31648133	$9.5068649095 \cdot 10^{-4}$



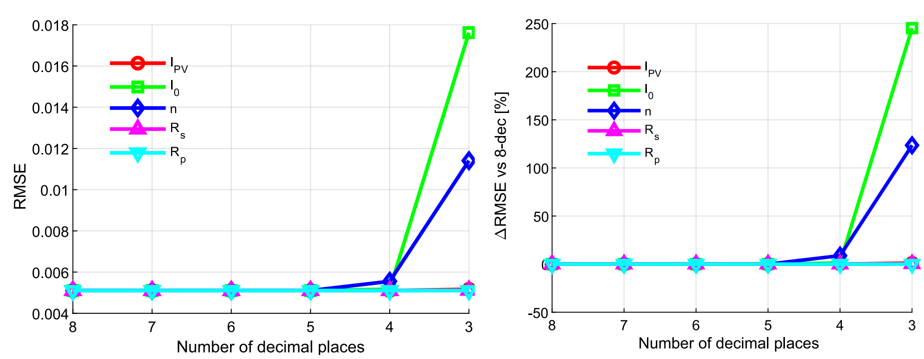
a) b)

Fig. 7. Impact of parameter truncation on SDM accuracy for the CdTe75669 solar module a) RMSE vs. number of decimals, b) relative RMSE increase (ERWCA).

Table 10
The estimated and literature-reported SDM parameter values for the KC200GT solar module.

Ref.	Method	I_{PV} [A]	I_0 [μA]	n	R_S [Ω]	R_P [Ω]
Estimated values in this study	ERWCA*	8.21192233	0.05752541	1.26338213	0.23712363	373.47451456
[40]	A&I	8.139	3e-04	1	0.269	166.7
[47]	**	8.201	0.1781	1.334	0.2201	951.93
[48]	BFPA	8.21502928	0.00289278	1.08890343	0.34006651	895.87425192
[49]	WHHO	8.21860582	1.43601e-3	1.05528589	0.24093983	774.212315
[50]	IELA	8.2183	9.6575	1.738	0.0169	765.034
Literature range		8.139–8.21860582	3e-04–9.6575	1–1.738	0.0169–0.34006651	166.7–951.93
Comment		within the reported range	within the reported range	within the reported range	within the reported range	within the reported range

* The parameter values marked with an asterisk represent the reference optimized SDM parameter sets used in the subsequent decimal-place truncation, sensitivity, and pairwise interaction analyses.
** not defined.



a) b)

Fig. 8. Impact of parameter truncation on SDM accuracy for the KC200GT solar module a) RMSE vs. number of decimals, b) relative RMSE increase (ERWCA).

information: the first indicator represents the actual increase in error, whereas the second indicates the sensitivity of the error with respect to the relative change in the parameter itself.

The parameter R_P shows the smallest influence on model accuracy in all analyzed cases. Its decimal-place truncation leads to negligible changes in the RMSE value and very small $|S_p|$ values. Therefore, it can be concluded that R_P does not require the same level of numerical precision as the parameters n, I_{PV} and I_0.

Overall, the results presented in Tables A4, B4, C4, D3 and E3 and in Fig. 9 confirm that the required numerical precision is not the same for all SDM parameters. The greatest attention should be paid to the diode ideality factor n, followed by the parameters I_{PV} and I_0, whereas R_P can be represented with fewer decimal places without significantly affecting model accuracy.

This table provides a practical guideline for determining the meaningful numerical precision for each parameter. Users of PV modeling software can directly refer to these results to select the appropriate number of significant digits, ensuring sufficient model accuracy without

Table 11
Relative change in RMSE (%) and absolute normalized sensitivity coefficient |Sp| with respect to the full-precision reference case for all considered solar cells and modules.

Parameter	Decimal places	RTC France		MSX-60		PWP 201		CdTe75669		KC200GT	
		ΔRMSE/ $RMSE_0$	$\|S_p\|$	ΔRMSE/ $RMSE_0$	$\|S_p\|$	ΔRMSE/ $RMSE_0$	$\|S_p\|$	ΔRMSE/ $RMSE_0$	$\|S_p\|$	ΔRMSE/ $RMSE_0$	$\|S_p\|$
I_{PV}	**7**	0.000000303	0.038346332	0.000000771	0.367257417	0.0000000679	0.007788493	0.0000000351	0.020387068	0.0000000030	0.008309741
	6	0.000062410	0.49458814	0.000009672	0.376113733	0.0000028955	0.050663563	0.0000392360	0.495727173	0.0000001033	0.025709415
	5	0.004230174	4.043047447	0.000042362	0.405633846	0.0004785605	0.650916481	0.0001691334	1.023880631	0.0000071677	0.252620266
	4	0.514307057	44.48369902	0.002124544	1.094428465	0.0275454040	4.937785537	0.1224470602	27.41486418	0.0006857154	2.521738358
I_0	**7**	0.000000069	0.005276201	0.000001738	0.209575853	0.0000000015	0.000637678	0.0000000008	not defined*	0.0000001749	0.010060943
	6	0.000000069	0.005276201	0.000079562	0.188098697	0.0000000832	0.003145557	0.0000000078	0.000313797	0.0003296937	0.462579577
	5	0.000662203	0.509247059	0.000172435	0.059233903	0.0000054525	0.024050616	0.0000121410	0.002563149	0.0575334352	6.117623752
	4	0.286059662	10.57522268	0.179604296	2.945928492	0.0015339810	0.400348144	0.0005502854	0.146431892	1.2621040904	28.57263096
n	**7**	0.000000035	0.052344143	0.000019039	3.593158771	0.0000000769	0.033759962	0.0000001351	0.06579469	0.0000012965	0.545991036
	6	0.000141102	3.417158535	0.000254660	4.369221126	0.0000273415	0.679218018	0.0000188379	0.679848759	0.0000233647	2.270653206
	5	0.021980996	42.66992718	0.005647864	11.02123417	0.0041783604	8.424716869	0.0056981679	11.73274057	0.0061980362	36.76285556
	4	1.720991961	376.0339714	0.808866977	110.425912	0.0689823736	34.23446028	0.6166348772	121.9136684	8.8234576221	1357.287067
R_S	**7**	0.000000364	0.002653212	0.000000876	0.039900334	0.000000000045	2.8021E-05	0.0000000008	0.000671773	0.0000000659	0.005208374
	6	0.000133060	0.051188509	0.000002615	0.039729058	0.0000000068	0.000261952	0.0000000016	0.000124508	0.0000100588	0.037860147
	5	0.007126744	0.374763701	0.000070417	0.031156432	0.0000034143	0.005786346	0.0000001060	0.001936215	0.0003078773	0.201115662
	4	0.324865534	2.528827358	0.002050543	0.071718705	0.0001424867	0.037354499	0.0000526271	0.050023055	0.0128494724	1.289425957
R_P	**7**	0	8.24244E-06	0.000000000038	0.016610006	0.0000000000039	0.000436349	0.0000000000	0.003044182	0.0000000001	0.003364322
	6	0	5.83695E-05	0.000000000643	0.017962702	0.0000000000039	3.2E-05	0.0000000000	0.000389012	0.0000000001	0.000462821
	5	0	5.59145E-05	0.000000016852	0.018113133	0.0000000000020	7.57145E-06	0.0000000000	0.000258227	0.0000000003	0.000235425
	4	0.000000019	0.000111485	0.000000037114	0.018117197	0.0000000000216	3.83751E-06	0.0000000013	0.000266983	0.0000000008	0.000197339

unnecessary computational effort.

* In this case, decimal-place truncation does not change the parameter value, i.e., $\Delta p / p_0 = 0$, so the normalized sensitivity coefficient |Sp| cannot be calculated.

3.7. Controlled percentage perturbation

The sensitivity of the SDM to variations in individual parameters was further examined through a controlled percentage parameter perturbation analysis. Unlike the previous analysis, where parameter changes resulted from decimal-place truncation of their values, in this approach each parameter was varied by a predefined relative amount. This enabled a more direct comparison of the influence of individual parameters on model accuracy.

For each considered SDM parameter, its value was individually increased and decreased with respect to the reference value p_0, while all other parameters were kept at their initial optimized values. The percentage variation of the parameters was performed with a step of 0.1 %, covering a wider range of positive and negative deviations from the reference value. For each new parameter value, the I–V characteristic was recalculated, and the corresponding change in RMSE was then determined with respect to the reference case.

A graphical representation of the obtained results is shown in Fig. 10, where the values of the absolute normalized sensitivity coefficient |Sp| and the relative error change ΔRMSE/$RMSE_0$ are presented as functions of the percentage parameter perturbation for all analyzed cells and modules. The perturbations were considered in the range from −5 % to + 5 %, while the zero value was omitted because it corresponds to the reference case without parameter perturbation. The tabulated results for selected characteristic perturbation points, −5 %, −1 %, −0.1 %, 0.1 %, 1 %, and 5 %, are provided in the Appendix, in Tables A5, B5, C5, D4, and E4.

The obtained results confirm the conclusions of the previous analysis based on decimal-place parameter truncation. In all analyzed cases the diode ideality factor n shows the highest sensitivity to controlled percentage variation. Even small perturbations of this parameter lead to a pronounced increase in the RMSE value, indicating that the model accuracy is particularly dependent on the precise determination and representation of the ideality factor.

For the RTC France cell, the parameter n has the strongest influence on RMSE, followed by I_{PV}, while I_0, R_S, and especially R_P have a weaker influence. A similar trend is observed for the MSX- 60 module, where n clearly stands out as the dominant parameter, whereas R_P shows a negligible influence on the RMSE variation. For the Photowatt-PWP 201 and CdTe75669 modules, the results also confirm the dominant influence of n, with a significant contribution of I_{PV}, especially under larger positive and negative perturbations.

The parameter R_P shows the lowest sensitivity in all analyzed cases. Even for perturbations of ± 5 %, the change in the RMSE value for this parameter remains considerably smaller than the changes caused by perturbations of n, I_{PV} and I_0. This result is fully consistent with the previous decimal-place truncation analysis, where R_P also exhibited the highest numerical stability.

It is important to emphasize that controlled percentage perturbation represents a stricter and more direct sensitivity test than decimal-place truncation, because all parameters are varied by the same relative amount. Therefore, the results presented in Fig. 10 further confirm that the required numerical precision is not the same for all SDM parameters. The greatest attention should be paid to the diode ideality factor n, followed by the parameters I_{PV} and I_0, whereas R_P is the least critical parameter from the perspective of preserving model accuracy.

3.8. Noise effects

To address the influence of measurement noise on the interpretation of meaningful numerical precision, an additional Monte Carlo current-

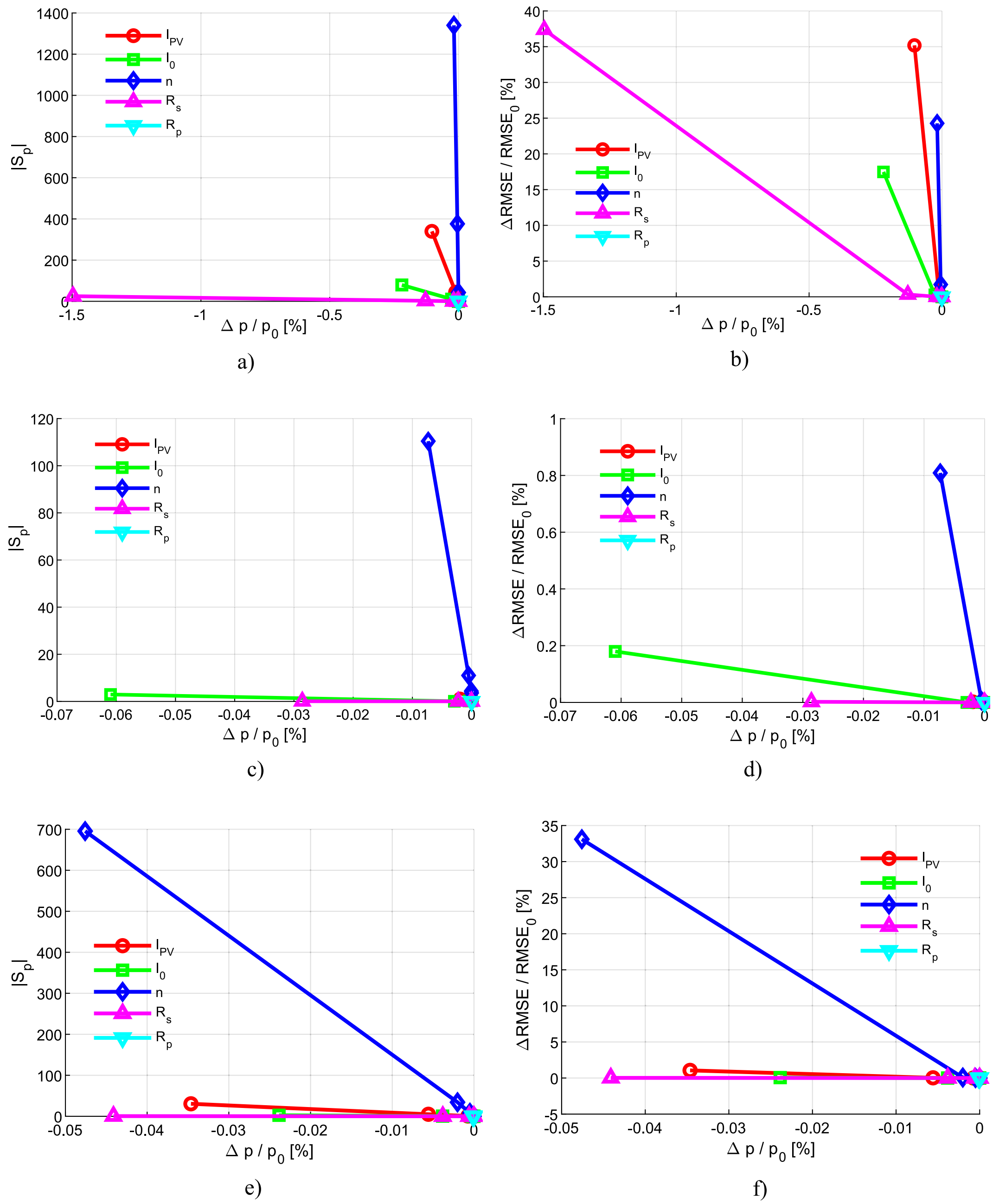


Fig. 9. A) rtc france solar cell: $|S_p|$ vs $\Delta p/p_0$ [%],b) RTC France solar cell: ΔRMSE/ $RMSE_0$ [%] vs $\Delta p/p_0$ [%]c) MSX-60 solar module: $|S_p|$ vs $\Delta p/p_0$ [%],d) MSX-60 solar module: ΔRMSE/ $RMSE_0$ [%] vs $\Delta p/p_0$ [%]e) Photowatt-PWP 201 solar module: $|S_p|$ vs $\Delta p/p_0$ [%],f) Photowatt-PWP 201 solar module: ΔRMSE/ $RMSE_0$ [%] vs $\Delta p/p_0$ [%]g) CdTe75669 solar module: $|S_p|$ vs $\Delta p/p_0$ [%],h) CdTe75669 solar module: ΔRMSE/ $RMSE_0$ [%] vs $\Delta p\ p_0$ [%]i) KC200GT: $|S_p|$ vs $\Delta p/p_0$ [%],j) KC200GT: ΔRMSE/ $RMSE_0$ [%] vs $\Delta p/p_0$ [%].

noise analysis was performed. Since the RMSE criterion used in this study is calculated from the residuals between the measured and modeled current values, the noise was introduced only into the measured current signal.

For each measured point *k,* the noisy current value is defined as:

$$I_{noisy}(k) = I_{meas}(k) + \Delta I(k) \tag{8}$$

where $\Delta I(k)$ represents a random current perturbation. In this analysis, the perturbation was generated as a uniformly distributed random value with respect to the short-circuit current:

$$\Delta I(k) = \alpha I_{SC}(2r_k - 1), \quad r_k \in [0,1] \tag{9}$$

where r_k is a random number from the interval [1], and α defines the maximum noise amplitude.

In the noise analysis, I_{SC} was approximated by the maximum measured current value, since the available experimental datasets contain the measured I–V points rather than explicit instrument uncertainty specifications:

$$I_{SC} \approx \max(I_{meas}) \tag{10}$$

In this way, the random perturbation for each dataset is scaled according to its own current level, which enables a comparable analysis for solar cells and modules with different nominal current values.

Since the original acquisition-device specifications are not available

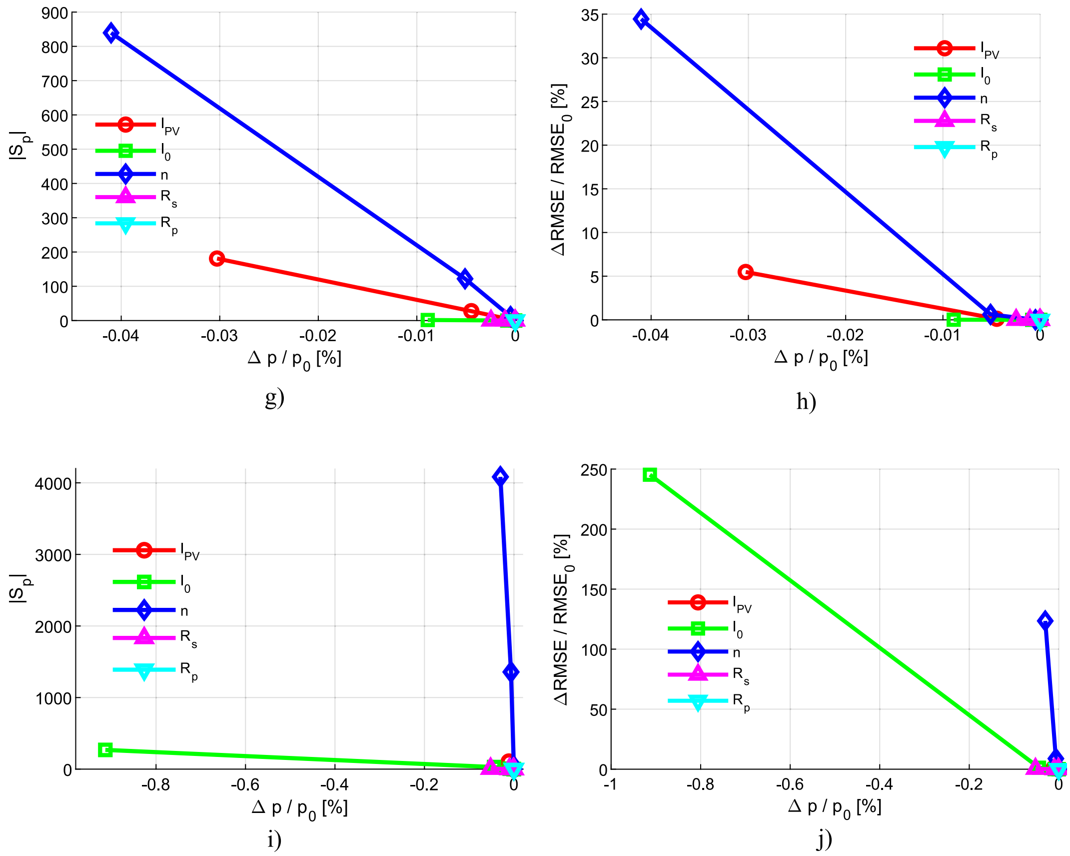


Fig. 9. (*continued*).

for the benchmark datasets, these noise levels should not be interpreted as exact instrument accuracies, but as representative reference levels for possible current-measurement uncertainty.

For each noise realization, the RMSE value was recalculated:

$$RMSE_{noise} = \sqrt{\frac{1}{N}\sum_{k=1}^{N}\left(I_{calc}(k) - I_{noisy}(k)\right)^2} \tag{11}$$

Three noise levels were considered in the Monte Carlo procedure:

$$\alpha = \{0.0005,\ 0.001,\ 0.005\} \tag{12}$$

which corresponds to maximum random perturbation amplitudes of $\pm$ 0.05 %, $\pm$0.1 % and $\pm$ 0.5 % with respect to I_{SC}. In other words, for each noise level, the following holds:

$$\Delta I \in [-\alpha I_{SC}, +\alpha I_{SC}] \tag{13}$$

For each noise level, 10,000 random realizations were generated, and the RMSE value was recalculated for each realization. In this way, it was estimated how much the RMSE can change solely due to possible measurement noise, independently of parameter truncation.

The obtained results, shown in Fig. 11(a)–(e), provide a reference for interpreting whether small RMSE changes caused by parameter truncation are practically meaningful or whether they may lie within the expected range of measurement uncertainty. Based on this analysis, RMSE changes caused by decimal-place truncation were not interpreted solely as numerical variations, but were compared with the RMSE variations that may arise from the assumed current-measurement noise. If a truncation-induced RMSE change is smaller than or comparable to the RMSE change produced by the Monte Carlo noise analysis, it is considered numerically detectable but not necessarily practically meaningful. Conversely, if the truncation-induced RMSE change clearly exceeds the noise-induced RMSE variation, it is considered practically meaningful. In this way, the proposed meaningful precision is evaluated not only according to whether the RMSE changes numerically, but also according to whether this change is significant relative to a reasonable estimate of the current-measurement uncertainty in the experimental I–V data.

It should be emphasized that the Monte Carlo simulation is used here only as a simple statistical robustness check, and not as a new optimization method or a new parameter-identification analysis. In the Monte Carlo procedure, a large number of random realizations of the measured current are generated for a given noise level, where the noise is uniformly distributed within a predefined interval. For each realization, the RMSE is calculated, after which the mean value and the range of the obtained RMSE values are examined. The aim is only to estimate the order of magnitude of the RMSE change that may occur due to possible measurement noise.

4. Joint parameter reduction: two-parameter interactions

In the previous analysis, the focus was on the individual impact of reducing each SDM parameter on model accuracy. However, in practical applications, multiple parameters may be reduced simultaneously, for example, due to limitations in data transmission, memory constraints, or the use of simplified model implementations. Such combined approximations may lead to nonlinear interactions among the parameters and can potentially increase the cumulative modelling error. Therefore, it is important to examine not only the individual influence of each parameter, but also the combined effect of simultaneous parameter reductions.

These findings are consistent with a basic principle of numerical analysis: parameters with different magnitudes and different physical roles in the model do not exhibit the same sensitivity to numerical precision loss. Parameters such as I_{PV}, R_S, n, and I_0 have a stronger influence on the calculated I–V curve, whereas R_P, whose value is typically

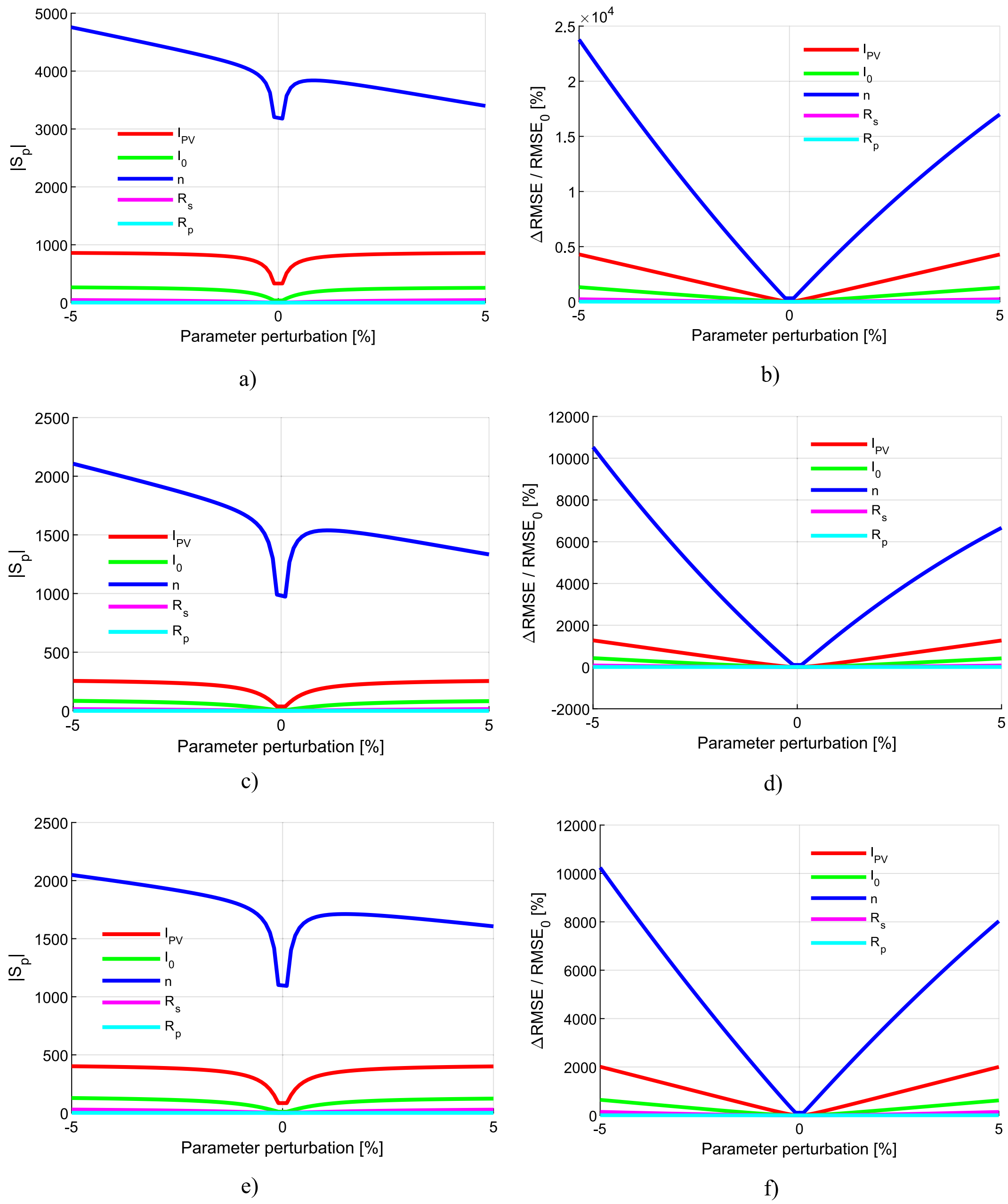


Fig. 10. Normalized sensitivity of SDM accuracy to controlled percentage perturbations of model parameters. a) RTC France solar cell using WaOA: $|S_p|$ vs. parameter perturbation [%],b) RTC France solar cell using WaOA: $\Delta RMSE/RMSE_0$ [%] vs. parameter perturbation [%],c) MSX-60 solar module using AWaOA: $|S_p|$ vs. parameter perturbation [%],d) MSX-60 solar module using AWaOA: $\Delta RMSE/RMSE_0$ [%] vs. parameter perturbation [%],e) Photowatt-PWP 201 solar module using CGTO: $|S_p|$ vs. parameter perturbation [%],f) Photowatt-PWP 201 solar module using CGTO: $\Delta RMSE/RMSE_0$ [%] vs. parameter perturbation [%],g) CdTe75669 solar module using ERWCA: $|S_p|$ vs. parameter perturbation [%],h) CdTe75669 solar module using ERWCA: $\Delta RMSE/RMSE_0$ [%] vs. parameter perturbation [%],i) KC200GT solar module using ERWCA: $|S_p|$ vs. parameter perturbation [%],j) KC200GT solar module using ERWCA: $\Delta RMSE/RMSE_0$ [%] vs. parameter perturbation [%],

much larger, remains relatively insensitive within the considered precision range.

The results of the previous individual reduction analysis provide the starting point for the present two-parameter study. They showed which parameters are individually sensitive to precise reduction, but they did not answer whether the simultaneous reduction of two parameters produces an independent, amplified, or partially compensated effect on the RMSE. This motivates the extended analysis presented in this section.

When the parameters of the single-diode model are properly estimated, the resulting current–voltage characteristic closely follows the measured experimental data. Any reduction in individual parameter values alters the shape of the calculated I–V curve and consequently increases its deviation from the measured characteristic. However, when two parameters are reduced simultaneously, their combined effect does not necessarily lead only to error amplification. Depending on the physical role of the parameters in the model, their effects may accumulate, but in certain cases they may also partially compensate each other, so that the overall model deviation remains small or even smaller than in the case of individual parameter reduction. Therefore, the

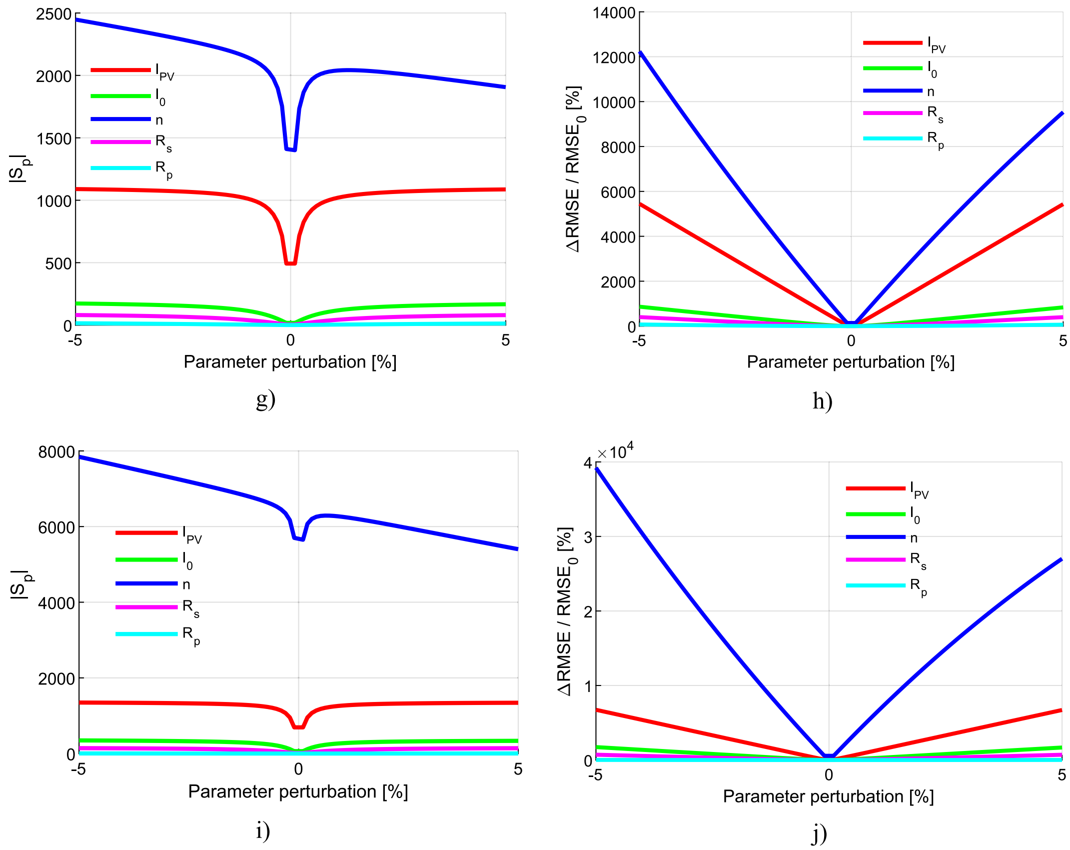


Fig. 10. (*continued*).

analysis of joint parameter reduction provides deeper insight into parameter interdependence and into the mechanism of error compensation. In this context, the term compensation does not imply that the model error disappears, but that the joint RMSE variation is smaller than the sum of the RMSE variations caused by the two parameters individually.

A reduction in the photogenerated current I_{PV} lowers the output current level, especially near the short-circuit region. In contrast, a reduction in the diode saturation current I_0 mainly affects the diode-controlled part of the model, particularly at higher voltages near the open-circuit condition, thereby changing the curvature of the I–V characteristic and the position of its knee. A decrease in the diode ideality factor n also modifies the exponential diode term and strongly influences the shape of the curve around the knee and in the high-voltage region, because n governs the sharpness of the transition between the nearly constant-current region and the drop toward open-circuit voltage. A reduction in the series resistance R_S decreases the internal voltage drop associated with the output current, making the modeled curve appear more ideal and less affected by current-induced voltage loss, with the strongest influence observed in the medium- and high-current regions, particularly around the maximum power point. Finally, a reduction in the shunt resistance R_P increases leakage current through the parallel branch and therefore primarily affects the slope of the curve near short-circuit conditions. When R_P is already very large, small variations in its value usually have only a weak influence, which explains why this parameter often exhibits the lowest sensitivity.

When two parameters are simultaneously reduced, their joint influence on the I–V characteristic depends on the physical role that each of them has in the single-diode model. In some cases, both parameters affect the same region of the curve in a similar direction, so their effects accumulate and lead to a larger deviation from the measured characteristic. In other cases, the influence of one parameter may partially counterbalance the effect of the other, resulting in partial error compensation and a smaller increase in RMSE than might be expected from the individual reductions. This behavior reflects the intrinsic interdependence of model parameters and explains why joint parameter reduction cannot always be interpreted as a simple superposition of individual effects. This is particularly important when the reduction level is small. As shown in the individual results, small reductions in precision often cause only minor RMSE changes. In that range, the SDM behaves almost locally linearly. However, when the reduction becomes stronger, nonlinear terms, especially the exponential diode term, become more influential, and the same parameter pair may exhibit either compensation or additional error amplification. Therefore, the interaction behavior must be interpreted for each reduction level, rather than as a fixed property of a parameter pair.

The interaction effects can be interpreted from the explicit form of the SDM current expressed using the Lambert W function, as given in Eqs. (2) and (3). This expression shows that the parameters do not affect the current independently: some parameters appear in the quasi-linear part, while others are coupled within the nonlinear argument of the Lambert W function and its exponential term.

In order to further verify the consistency of the obtained sensitivity results, an additional pairwise interaction analysis was performed. While the previous analysis considered the influence of each SDM parameter separately, this analysis examines the effect of the simultaneous reduction of two parameters. The reductions were performed for three levels: 1 %, 2 %, and 5 %.

For each considered parameter pair and each reduction level, three RMSE values were calculated: $RMSE_i$, obtained when only the first parameter was reduced; $RMSE_j$, obtained when only the second parameter was reduced; and $RMSE_{ij}$, obtained when both parameters were reduced simultaneously. The corresponding relative RMSE variations were defined as:

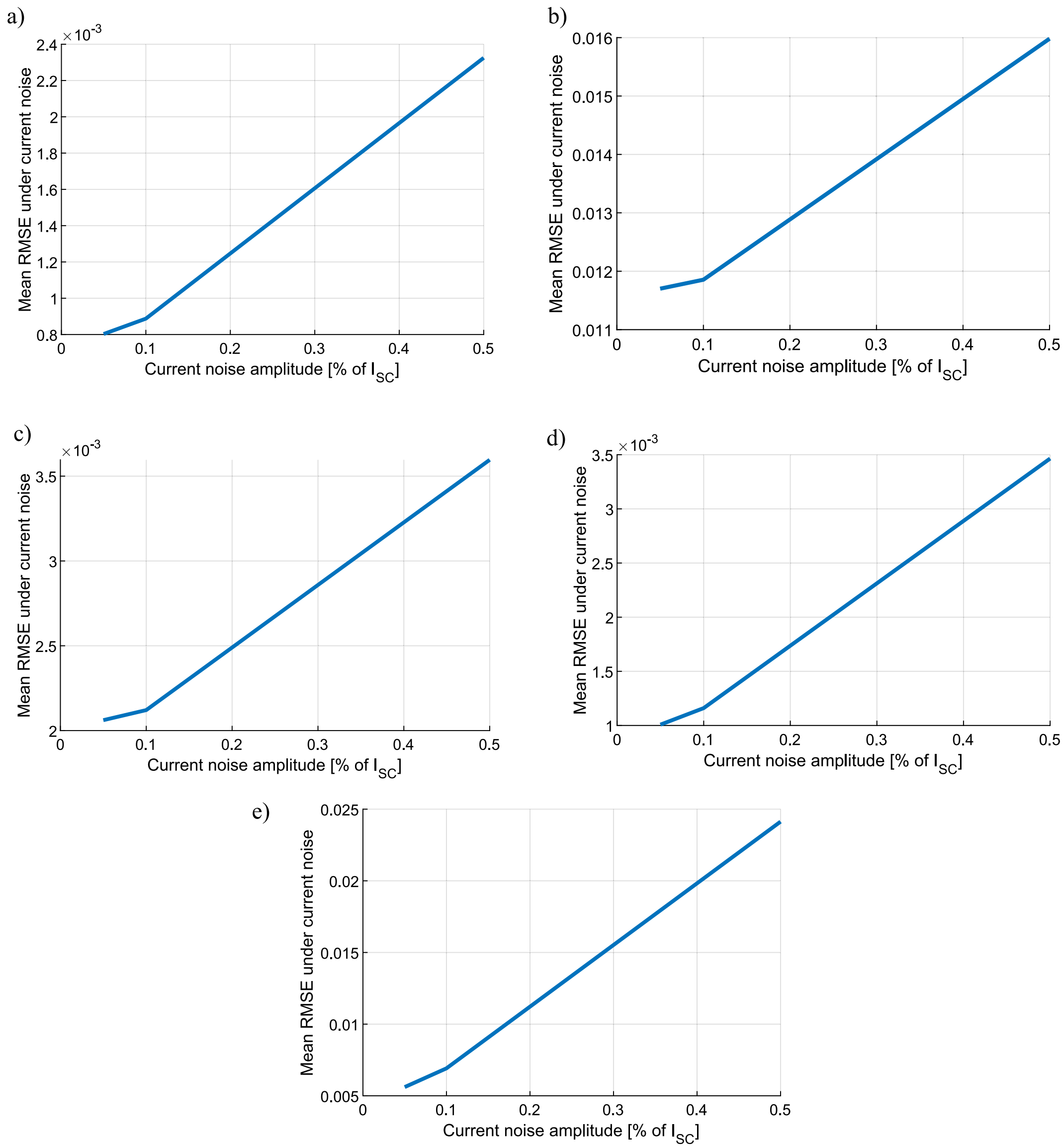


Fig. 11. Monte Carlo assessment of current-measurement noise influence on RMSE for: a) RTC France solar cell (WaOA), b) MSX-60 solar module (AWaOA), c) Photowatt-PWP 201 solar module (CGTO), d) CdTe75669 solar module (ERWCA), and e) KC200GT solar module (ERWCA).

$$E_i = \frac{RMSE_i - RMSE_0}{RMSE_0} \cdot 100 \tag{14}$$

$$E_j = \frac{RMSE_j - RMSE_0}{RMSE_0} \cdot 100 \tag{15}$$

$$E_{ij} = \frac{RMSE_{ij} - RMSE_0}{RMSE_0} \cdot 100 \tag{16}$$

where $RMSE_0$ is the reference RMSE obtained using the original optimal parameter set.

The interaction index was then calculated as:

$$I_{ij} = E_{ij} - \left(E_i + E_j\right) \tag{17}$$

The interaction term was used to determine whether the combined influence of two parameters is approximately additive, cumulative, or compensating. A negative value of I_{ij} indicates that the joint effect of reducing two parameters is smaller than the sum of their individual effects, corresponding to partial compensation. A positive value of I_{ij} indicates that the joint effect is larger than the sum of the individual effects, corresponding to additional error amplification. Values close to zero indicate approximately additive behavior.

For each parameter pair, the individual RMSE variations E_i and E_j, the combined RMSE variation E_{ij}, and the interaction term I_{ij} were calculated. The corresponding numerical results for all analyzed cells and modules are presented in Fig. 12, while the complete tabulated data are provided in Appendix F, Tables F1 - F5.

In the following, each parameter pair is analyzed through two aspects:

- a mathematical analysis of the mutual dependence between the considered parameters and

- an analysis of the results based on results presented in the Tables G1 – G5 (see Appendix G). A detailed description of this analysis is provided in Table 12.

5. Fixing the parallel resistance: is high precision really needed?

In the previous analyses, it has been shown that changing the number of significant digits of the shunt resistance R_P has a very small impact on the overall accuracy of the single-diode model. Unlike the other parameters, reducing the numerical precision of R_P has almost no effect on the RMSE, both in the case of individual truncation and in joint

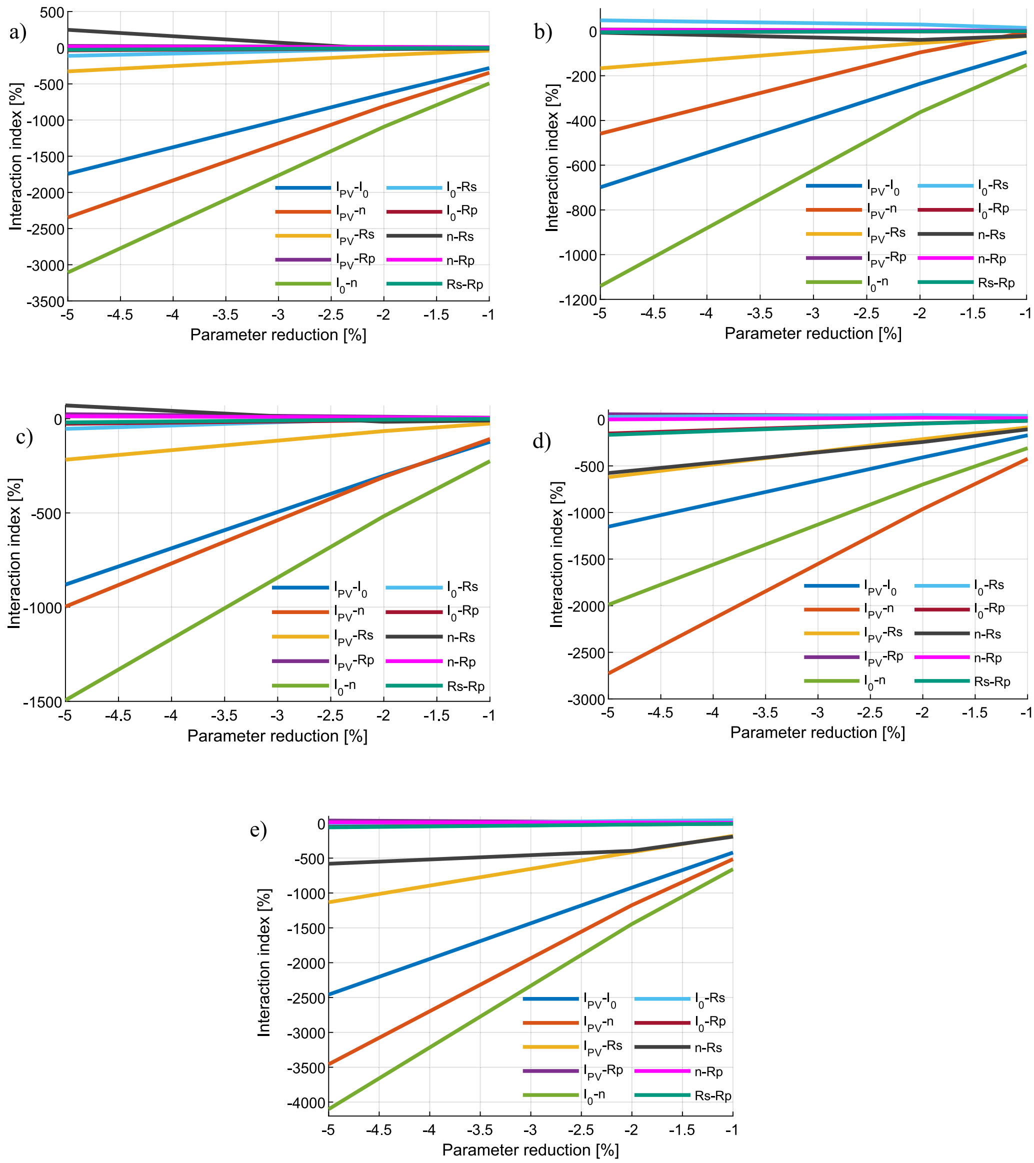


Fig. 12. Pairwise interaction indices under controlled SDM parameter reductions for the considered solar cells and modules: a) RTC France solar cell (WaOA), b) MSX-60 solar module (AWaOA), c) Photowatt-PWP 201 solar module (CGTO), d) CdTe75669 solar module (ERWCA), and e) KC200GT solar module (ERWCA).

truncation with other parameters. This naturally raises the question of whether it is necessary to retain a high numerical precision for R_P, or whether this parameter can be fixed to a coarser value without degrading the quality of the model.

For this purpose, this section examines the procedure of fixing the shunt resistance to a prescribed value, while the remaining four SDM parameters are re-estimated with full numerical precision. This approach makes it possible to assess:

- how changes in the numerical precision of R_P are reflected in the newly obtained values of the remaining parameters,
- the extent to which the model is sensitive to fixing R_P to a prescribed value,
- the resulting difference in RMSE with respect to the reference case in which all parameters are optimized without constraints.

5.1. RTC France solar cell

Table 13 presents the re-estimated SDM parameter values for the RTC France solar cell when the parallel resistance R_P is fixed at predefined values ranging from 8 to 3 decimal places. The remaining four parameters (I_{PV}, I_0, n and R_S) are optimized using the WaOA algorithm. The table also reports the resulting RMSE for each precision level of R_P. As can be seen from the table, the RMSE remains virtually unchanged, whereas the parameter values exhibit noticeable variations in the less significant decimal places.

Fig. 13 illustrates the impact of fixing the shunt resistance on the I–V and P–V characteristics of the RTC France solar cell. Fig. 13a shows the modelled I–V characteristics obtained for different precisions of the fixed R_P (from 8 down to 3 decimal places), while Fig. 13b presents the corresponding current deviation with respect to the reference case with full precision. Fig. 13c displays the P–V characteristics for the same precision levels, and Fig. 13d shows the corresponding power deviation relative to the reference case.

The results show that fixing the shunt resistance at different levels of parameter truncation has practically no effect on the I–V characteristics of the RTC France solar cell. As shown in Fig. 13a, all modelled curves remain virtually identical regardless of whether R_P is specified with 8 or only 3 decimal places. The absence of any visible deviation is confirmed

Table 12
Summary of mathematical interpretation and numerical results for pairwise SDM parameter interactions.

Pair	Mathematical interpretation	Analysis of the numerical results
I_{PV}-n	For the I_{PV}-n parameter pair, the explicit Lambert W expression for the current, given in Eqs. (2) and (3), shows that these two parameters appear together in the same nonlinear part of the SDM model. The parameter I_{PV} primarily affects the quasi-linear part of the current through the term $R_P(I_{PV}+I_0)/(R_S+R_P)$, but it also appears in the numerator of the exponential argument through $V+R_S(I_{PV}+I_0)$. On the other hand, the diode ideality factor n appears in the denominator of the same exponential argument, as well as in front of the Lambert W function. Therefore, a reduction in I_{PV} tends to decrease the exponential argument, whereas a reduction in n tends to increase it. These opposite effects within the same nonlinear term prevent the individual effects from being fully additive, resulting in a subadditive, i.e., compensating response.	The obtained results for the I_{PV}-n pair confirm the previous mathematical interpretation, since the interaction index I_{ij} is negative for all analyzed cells/modules and for all reduction levels. This means that, in each case, the joint error E_{ij} is lower than the sum of the individual errors E_i+E_j, so the interaction is classified as compensating. However, the values of E_{ij} are still very large, indicating that this parameter pair is highly sensitive under simultaneous reduction. For example, at a reduction level of −5 %, E_{ij} is 25735.62 % for RTC France and 42499.26 % for KC200GT, but in both cases it remains lower than the additive expectation. Therefore, a negative I_{ij} does not indicate a weak joint effect; rather, it shows that the strong individual effects of I_{PV} and n partially compensate each other within the nonlinear structure of the SDM model.
I_{PV}-I_0	For the I_{PV}-I_0 parameter pair, the explicit Lambert W expression for the current, given in Eqs. (2) and (3), shows that I_{PV} and I_0 appear together in the term $I_{PV}+I_0$, which affects both the quasi-linear part of the current and the exponential argument of the Lambert W function. When I_{PV} is reduced, the quasi-linear part of the current decreases, while the exponential argument is also reduced through the term $V+R_S(I_{PV}+I_0)$. Since the Lambert W function is monotonically increasing for positive arguments, a reduction in the argument x leads to a reduction in $W(x)$. Due to the negative sign in front of the Lambert W term, this nonlinear part partially mitigates the overall current change, so the effects of simultaneous reductions in I_{PV} and I_0 are not fully additive, but instead result in a subadditive, i.e., compensating response.	The results for the I_{PV}-I_0 pair in Tables G-1 – G-5 are consistent with the previous mathematical interpretation, since the values of the interaction index I_{ij} are negative for all analyzed cells/modules and for all reduction levels. This means that the joint RMSE variation E_{ij} is always lower than the sum of the individual variations E_i+E_j, so the interaction is classified as compensating. For example, for the RTC France cell at a reduction level of −5 %, the sum of the individual effects is 4296.29 + 1324.38 = 5620.67 %, whereas the joint effect is E_{ij} = 3877.39 %, giving I_{ij} = -1743.29 %. The same pattern is observed for the MSX-60, Photowatt-PWP 201, CdTe75669, and KC200GT modules, where all I_{ij} values for this pair are negative. Therefore, the simultaneous reduction of I_{PV} and I_0 has a significant effect on the RMSE, but this effect remains lower than the additive expectation based on their individual reductions.
I_0-n	For the I_0-n parameter pair, Eqs. (2) and (3) show that I_0 and n are directly connected within the nonlinear diode-related part of the SDM model. The parameter I_0 appears in the multiplicative factor of the Lambert W argument and also in the exponential argument through the term $I_{PV}+I_0$. On the other hand, n appears in front of the Lambert W function and in the denominator of the argument x , as well as in the denominator of the exponential term. When I_0 is reduced, the multiplicative factor of the argument x decreases, and the exponential argument is also partially reduced through the term $V+R_S(I_{PV}+I_0)$. Since the Lambert W function is monotonically increasing for positive arguments, a reduction in x leads to a reduction in $W(x)$. Due to the negative sign in front of the Lambert W term, the reduction in $W(x)$ mitigates the negative diode contribution to the current. However, when n is reduced, the denominator in the argument x decreases, which tends to increase the exponential argument, while the factor nV_{th}/R_S in front of the Lambert W function also decreases. Because of these opposite and mutually connected effects of I_0 and n within the same nonlinear diode term, their effects are not fully additive under simultaneous reduction, leading to a subadditive, i.e., compensating response.	The results for the I_0-n pair in Tables G-1 – G-5 are consistent with the previous mathematical interpretation, since the values of the interaction index I_{ij} are negative for all analyzed cells/modules and for all reduction levels. This means that the joint RMSE variation E_{ij} is always lower than the sum of the individual variations E_i+E_j, so the interaction is classified as compensating. For example, for the RTC France cell at a reduction level of −5 %, the sum of the individual effects is 1324.38 + 23785.54 = 25109.92 %, whereas the joint effect is E_{ij} = 22002.24 %, giving I_{ij} = -3107.69 %. The same pattern is observed for the MSX-60, Photowatt-PWP 201, CdTe75669, and KC200GT modules, where all I_{ij} values for this pair are negative. Therefore, although the joint RMSE effect of the I_0-n pair is very pronounced due to the dominant influence of the diode ideality factor n, this effect remains lower than the additive expectation, indicating that the reductions in I_0 and n partially compensate each other within the nonlinear diode term of the SDM model.
I_{PV}-R_S	For the I_{PV}-R_S parameter pair, it follows from Eqs. (2) and (3) that I_{PV} primarily affects the quasi-linear part of the current through the term $R_P(I_{PV}+I_0)/(R_S+R_P)$, but it also appears in the exponential argument through $V+R_S(I_{PV}+I_0)$. The parameter R_S is more deeply involved in the structure of the expression, since it appears in the denominator of the quasi-linear term, in the coefficient nV_{th}/R_S in front of the Lambert W function, in the multiplicative factor of the argument x , and in the exponential argument. When I_{PV} is reduced, the quasi-linear part of the current decreases, while the exponential argument of the Lambert W function is also reduced. When R_S is reduced, both the quasi-linear and nonlinear parts of the expression are modified: the series voltage drop decreases, the argument x changes, while the factor nV_{th}/R_S in front of the Lambert W function increases. Therefore, the effect of reducing R_S is not simply added to the effect of reducing I_{PV}, but depends on the balance of several terms in the Lambert W expression. Since both parameters affect the same nonlinear mechanism through the term $V+R_S(I_{PV}+I_0)$, their effects partially overlap under simultaneous reduction, leading to a subadditive, i.e., compensating response.	The results for the I_{PV}-R_S pair in Tables G-1 – G-5 support this interpretation, since the interaction index I_{ij} is negative for all analyzed cells/modules and for all reduction levels. Thus, the joint RMSE variation E_{ij} is always lower than the sum of the individual variations E_i+E_j, and the interaction is classified as compensating. For example, for the RTC France cell at a reduction level of −5 %, the sum of the individual effects is 4296.29 + 231.99 = 4528.28 %, whereas the joint effect is E_{ij} = 4201.00 %, giving I_{ij} = -327.29 %. A consistent trend is also observed for the MSX-60, Photowatt-PWP 201, CdTe75669, and KC200GT modules, where all I_{ij} values for this pair are negative. Therefore, the simultaneous reduction of I_{PV} and R_S affects the RMSE, but the effect remains lower than the additive expectation because their influences partially overlap through the quasi-linear part and the nonlinear Lambert W term of the SDM model.
n-R_S	For the n-R_S parameter pair, Eqs. (2) and (3) indicate a strong mutual dependence within the nonlinear part of the SDM current expression. The diode ideality factor n appears both in the prefactor nV_{th}/R_S of the Lambert W term and in the denominator of the argument x , including the denominator of the exponential argument. The parameter R_S is also involved in several parts of the expression: in the denominator of the quasi-linear term, in the prefactor nV_{th}/R_S, in the	The results for the n-R_S pair show that this interaction is not fully uniform across all cases, which agrees with the mathematical structure of the SDM current expression. For the RTC France and Photowatt-PWP 201 cases at the −5 % reduction level, the I_{ij} values are positive, 247.31 % and 70.38 %, respectively, indicating a cumulative interaction. This means that, under stronger reduction, the joint RMSE effect is larger than the sum of the individual effects, so the

(continued on next page)

Table 12 (*continued*)

Pair	Mathematical interpretation	Analysis of the numerical results
	multiplicative factor of x , and in the exponential argument through $V + R_S(I_{PV} + I_0)$. Therefore, reductions in n and R_S modify the same nonlinear Lambert W contribution, but through different and partly opposing mechanisms. When n is reduced, the prefactor nV_{th}/R_S decreases, while the argument x tends to increase because n appears in the denominator of both the multiplicative and exponential parts. In contrast, reducing R_S increases the prefactor nV_{th}/R_S, but also changes the multiplicative factor of x , the exponential argument, and the quasi-linear part of the current. As a result, the combined effect of reducing n and R_S cannot be described in advance as strictly compensating or strictly cumulative. Instead, their interaction depends on which part of the Lambert W expression dominates for a given module and reduction level.	reductions in n and R_S additionally amplify the error. However, at lower reduction levels of -2 % and -1 %, I_{ij} becomes negative for these two cases, indicating a transition toward compensating behavior. For the MSX-60, CdTe75669, and KC200GT modules, all I_{ij} values are negative for the investigated reduction levels, so the interaction is predominantly compensating. Therefore, the n-R_S pair clearly demonstrates that the interaction effect depends on both the reduction level and the specific parameter values of the considered module: stronger reductions may produce cumulative error amplification in some cases, whereas smaller reductions and the remaining modules are dominated by a subadditive, i.e., compensating response.
I_{PV}-R_P	For the I_{PV}-R_P parameter pair, Eqs. (2) and (3) indicate that I_{PV} contributes to the quasi-linear current term, while it also enters the exponential argument through $V + R_S(I_{PV} + I_0)$. The parameter R_P appears in several parts of the current expression: in the quasi-linear term, in the denominator $R_S + R_P$, in the multiplicative factor of the argument x , and in the exponential argument. When I_{PV} is reduced, the basic current level decreases and the exponential argument is partly reduced. A reduction in R_P modifies the shunt-related component of the model and changes the ratio involving R_P and $R_S + R_P$, thereby affecting both the quasi-linear and nonlinear parts of the current. Since I_{PV} dominantly controls the current level, whereas R_P affects the losses through the parallel branch, their simultaneous reduction may add a small additional disturbance to the already pronounced effect of reducing I_{PV}, leading to a cumulative response.	The results for the I_{PV}-R_P pair in Tables G-1 – G-5 show a consistently cumulative interaction for all analyzed cells/modules and all reduction levels. In all cases, the I_{ij} values are positive, meaning that the joint RMSE variation E_{ij} is greater than the sum of the individual variations $E_i + E_j$. For example, for the RTC France cell at a reduction level of -5 %, the sum of the individual effects is $4296.29 + 8.71 = 4305.00$ %, whereas the joint effect is $E_{ij} = 4330.30$ %, giving $I_{ij} = 25.30$ %. A similar cumulative tendency is observed for the MSX-60, Photowatt-PWP 201, CdTe75669, and KC200GT modules, where I_{ij} values for this pair remain positive at all reduction levels. However, it should be emphasized that the individual effect E_j associated with R_P is very small compared with the effect E_i associated with I_{PV}, indicating that the total joint error is mainly governed by the reduction in I_{PV}, while R_P adds a relatively small but consistently cumulative contribution.
I_0-R_S	For the I_0-R_S parameter pair, the structure of Eqs. (2) and (3) indicates that both parameters are involved in the nonlinear Lambert W term. The parameter I_0 directly scales the argument x , and it also contributes to the exponential argument through the term $I_{PV} + I_0$. The series resistance R_S appears in the denominator of the quasi-linear term, in the prefactor nV_{th}/R_S, in the multiplicative factor of x , and in the exponential argument through $V + R_S(I_{PV} + I_0)$. Therefore, reductions in I_0 and R_S affect the same nonlinear diode-related contribution, but through different parts of the current expression, which makes their combined effect dependent on both the reduction level and the specific module parameters. When I_0 is reduced, the argument x decreases and, since the Lambert W function is monotonically increasing for positive arguments, $W(x)$ also decreases. Due to the negative sign in front of the Lambert W term, this effect partly mitigates the negative diode contribution to the current. Reducing R_S, however, modifies several terms at once: the prefactor nV_{th}/R_S increases, the ratio $R_S/(R_S + R_P)$ changes, and the exponential argument is also affected. As a result, the effects of reducing I_0 and R_S cannot be combined in a simple additive way, but depend on the balance between the multiplicative factor, the exponential argument, and the prefactor of the Lambert W function.	The results for the I_0-R_S pair show that this parameter combination does not exhibit fully uniform behavior across all tables Tables G-1 – G-5. For the RTC France and Photowatt-PWP 201 cases, the interaction is compensating at reduction levels of -5 % and -2 %, while it becomes slightly cumulative at -1 %. For the MSX-60 and CdTe75669 modules, the I_{ij} values are positive for all reduction levels, indicating that the joint RMSE variation is greater than the sum of the individual variations and that cumulative behavior dominates. In the KC200GT case, the interaction is compensating at -5 %, but becomes cumulative at the lower reduction levels of -2 % and -1 %. These results are consistent with the mathematical structure of the SDM current expression, since I_0 and R_S are involved in the same nonlinear Lambert W term through several mutually competing factors. Therefore, no single interaction type can be assigned to this pair for all cases; instead, the interaction depends on the reduction level and on the specific parameter values of the analyzed cell or module.
I_0-R_P	For the I_0-R_P parameter pair, Eqs. (2) and (3) indicate that these parameters are linked through both parts of the SDM current expression. The parameter I_0 directly scales the argument x , while it also contributes to the exponential argument through the term $I_{PV} + I_0$. The shunt resistance R_P appears in the quasi-linear term, in the denominator $R_S + R_P$, in the multiplicative factor of x , and in the exponential argument. When I_0 is reduced, the Lambert W argument decreases, and consequently $W(x)$ also decreases because the Lambert W function is monotonically increasing for positive arguments. A reduction in R_P changes the balance between the shunt and series resistance terms in several parts of the expression, although its effect is often limited when R_P is already large. Therefore, under simultaneous reduction of I_0 and R_P, the dominant effect of I_0 is partly mitigated through changes in the same Lambert W term, leading to a subadditive, i.e., compensating response.	The results for the I_0-R_P pair in Tables G-1 – G-5 indicate a consistently compensating interaction for all analyzed cells/modules and all reduction levels. In every case, the I_{ij} values are negative, meaning that the joint RMSE variation E_{ij} is lower than the sum of the individual variations $E_i + E_j$. For example, for the RTC France cell at a reduction level of -5 %, the sum of the individual effects is $1324.38 + 8.71 = 1333.09$ %, whereas the joint effect is $E_{ij} = 1296.38$ %, giving $I_{ij} = -36.71$ %. The same tendency appears for the MSX-60, Photowatt-PWP 201, CdTe75669, and KC200GT modules, where all I_{ij} values for this pair are negative. In addition, the individual effect of R_P is much smaller than that of I_0, showing that the joint error is mainly governed by the reduction in I_0, while the reduction in R_P contributes more weakly and leads to partial compensation through the common nonlinear mechanism.
n-R_P	For the n-R_P parameter pair, Eqs. (2) and (3) show that the two parameters are connected through the nonlinear part of the SDM current expression. The diode ideality factor n appears in the prefactor nV_{th}/R_S, as well as in the denominator of the argument x and the denominator of the exponential argument. The shunt resistance R_P is included in the quasi-linear term, the denominator $R_S + R_P$, the multiplicative factor of x , and the exponential argument. When n is reduced, the prefactor of the Lambert W term decreases, while the argument x tends to increase because n appears in the denominator. In contrast, reducing R_P modifies both the quasi-linear part and the Lambert W argument; however, its individual influence is usually weak when R_P has a large value. Therefore, the combined	The results for the n-R_P pair in Tables G-1 – G-5 indicate a predominantly cumulative interaction. For the RTC France, MSX-60, Photowatt-PWP 201, and KC200GT cases, all I_{ij} values are positive for the three reduction levels, meaning that the joint RMSE variation E_{ij} is greater than the sum of the individual variations $E_i + E_j$. For example, for the KC200GT module at a reduction level of -5 %, the sum of the individual effects is $39217.75 + 15.63 = 39233.38$ %, whereas the joint effect is $E_{ij} = 39247.98$ %, giving $I_{ij} = 14.60$ %. The only exception is the CdTe75669 module at the -5 % reduction level, where $I_{ij} = -1.80$ %, indicating a very weak compensating effect; however, at -2 % and -1 %, this pair shifts to cumulative behavior. Therefore, although the individual effect of R_P

(*continued on next page*)

Table 12 (*continued*)

Pair	Mathematical interpretation	Analysis of the numerical results
	reduction of n and R_P is mainly governed by the reduction in n, while R_P introduces a smaller additional contribution.	is much smaller than that of n, its simultaneous reduction with n usually adds a small but positive contribution to the total RMSE error, so the n-R_P pair can be described as predominantly cumulative.
R_S-R_P	For the R_S-R_P parameter pair, Eqs. (2) and (3) reveal that these two resistive parameters are interrelated through several terms of the SDM current expression. Both parameters appear in the common denominator $R_S + R_P$, while R_S also affects the prefactor nV_{th}/R_S of the Lambert W term, and R_P directly contributes to the quasi-linear part of the current. In addition, both R_S and R_P are included in the Lambert W argument, through both the multiplicative factor and the exponential term. A reduction in R_S modifies the series voltage drop and the nonlinear diode contribution, whereas a reduction in R_P changes the effect of the shunt branch, especially in the lower-voltage region. Since both parameters act through common resistance-dependent terms, such as $R_S + R_P$ and $R_SR_P/(R_S + R_P)$, their effects partially overlap under simultaneous reduction, so the combined response cannot be represented by a simple sum of the individual effects.	The results for the R_S-R_P pair in Tables G-1 – G-5 show a consistently compensating interaction for all analyzed cells/modules and all reduction levels. In all cases, the I_{ij} values are negative, which means that the joint RMSE variation E_{ij} is lower than the sum of the individual variations $E_i + E_j$. For example, for the RTC France cell at a reduction level of −5 %, the sum of the individual effects is 231.99 + 8.71 = 240.70 %, whereas the joint effect is E_{ij} = 212.67 %, giving I_{ij} = − 28.03 %. The same tendency is observed for the MSX-60, Photowatt-PWP 201, CdTe75669, and KC200GT modules, where all I_{ij} values for this pair are negative. Therefore, the simultaneous reduction of R_S and R_P leads to a subadditive response, since their effects partially overlap through the common resistance-dependent terms of the SDM model and the nonlinear Lambert W argument.

by the current difference plot in Fig. 13b: although small oscillations can be observed, they remain on the order of only a few mill amperes, which is negligible compared to the absolute value of the cell current. Furthermore, there is no noticeable divergence of the curves either in the short-circuit region or in the vicinity of the open-circuit voltage.

A similar behaviour is observed for the P–V characteristics in Fig. 13c, where the power curves practically coincide over the entire voltage range. The power differences shown in Fig. 13d remain very small and do not exhibit any systematic trend of increase with decreasing precision of R_P. It is important to note that the largest deviations occur in the region near the end of the I–V curve (high voltage, low current), where the model naturally becomes more sensitive to numerical perturbations; however, even in this region the deviations remain sufficiently small not to affect the physical interpretation or the determination of the maximum power point.

Overall, the results clearly show that changes in the numerical precision of the shunt resistance do not affect the reconstructed I–V and P–V characteristics to an extent that would be relevant for modelling. At the same time, the re-estimation of the remaining parameters exhibits complete stability, with no indication of effects or degradation of accuracy.

5.2. MSX-60 solar module

Table 14 summarizes the re-estimated SDM parameters for the MSX-60 solar module under the condition that the parallel resistance is fixed to predefined values with decimal precision ranging from 8 to 3 digits. For each imposed precision level, the remaining four parameters are normally optimized using the AWaOA algorithm. The table also includes the corresponding RMSE values, providing a direct indication of how the precision of the fixed R_P influences the overall fitting accuracy. The table shows that the RMSE is almost identical across all cases, while the parameter values differ in some of the decimal places.

Fig. 14 illustrates the impact of fixing the shunt resistance on the I–V and P–V characteristics of the MSX-60 solar module. Fig. 14a shows the modelled I–V characteristics obtained for different precisions of the fixed R_P (from 8 down to 3 decimal places), while Fig. 14b presents the corresponding current deviation with respect to the reference case with full precision. Fig. 14c shows the P–V characteristics for the same precision levels, and Fig. 14d illustrates the corresponding power deviations relative to the baseline case.

The results shown in Fig. 14 indicate that the precision of the fixed shunt resistance has an extremely small impact on the accuracy of the SDM for the MSX-60 solar module. The I–V characteristics in Fig. 14a practically coincide for all precision levels, indicating that changing the number of significant digits of R_P does not affect either the shape or the position of the curve within the operating range of the module. This is further confirmed by Fig. 14b, where the current deviations are very small and do not exhibit any trend of increase with decreasing precision.

A similar pattern is observed in the P–V characteristics shown in Fig. 14c, where the maximum power and the position of the MPP point remain almost unchanged, while Fig. 14d shows that the power deviations are negligible compared to the overall power level of the module.

In the case of the MSX-60 module, fixing R_P does not introduce any measurable degradation of the model, confirming that this parameter plays a secondary role in the overall numerical sensitivity of the SDM. This further consolidates the results obtained for the RTC cell and shows that the low influence of the shunt resistance is a robust observation that also holds for modules operating at higher voltage and current levels.

6. Discussion of results with limitations

The results clearly show that different SDM parameters exhibit markedly different sensitivities to parameter truncation and numerical precision reduction. The diode ideality factor n, the photogenerated current I_{PV}, and, in some cases, the diode saturation current I_0 and series resistance R_S, are the parameters for which reducing the number of significant digits below a certain threshold may lead to a substantial increase in the error between the model and the experimental data. This highlights their key role in accurately capturing the behavior of solar cells and PV modules and underscores the need for a higher level of precision in their representation.

By contrast, the shunt resistance R_P shows a relatively low impact on the overall error, even when its numerical precision is significantly reduced. Its decimal-place truncation produces negligible RMSE variations and very small normalized sensitivity coefficients. This suggests

Table 13
Re-estimated SDM parameters for the RTC France solar cell under fixed R_P.

Fixed R_P [Ω]	I_{PV} [A]	I_0 [μA]	n	R_S[Ω]	RMSE
52.88969619	0.76078796	0.310684142	1.47726764	0.03654695	$7.73006269004 \cdot 10^{-4}$
52.8896961	0.760787971	0.310684101	1.477267627	0.03654695	$7.73006269004 \cdot 10^{-4}$
52.889696	0.760787971	0.310684101	1.477267627	0.03654695	$7.73006269004 \cdot 10^{-4}$
52.88969	0.760787971	0.310684101	1.477267627	0.03654695	$7.73006269005 \cdot 10^{-4}$
52.8896	0.760787976	0.310683617	1.477267473	0.03654695	$7.73006269036 \cdot 10^{-4}$
52.889	0.760788007	0.310680575	1.477266498	0.03654699	$7.73006269725 \cdot 10^{-4}$

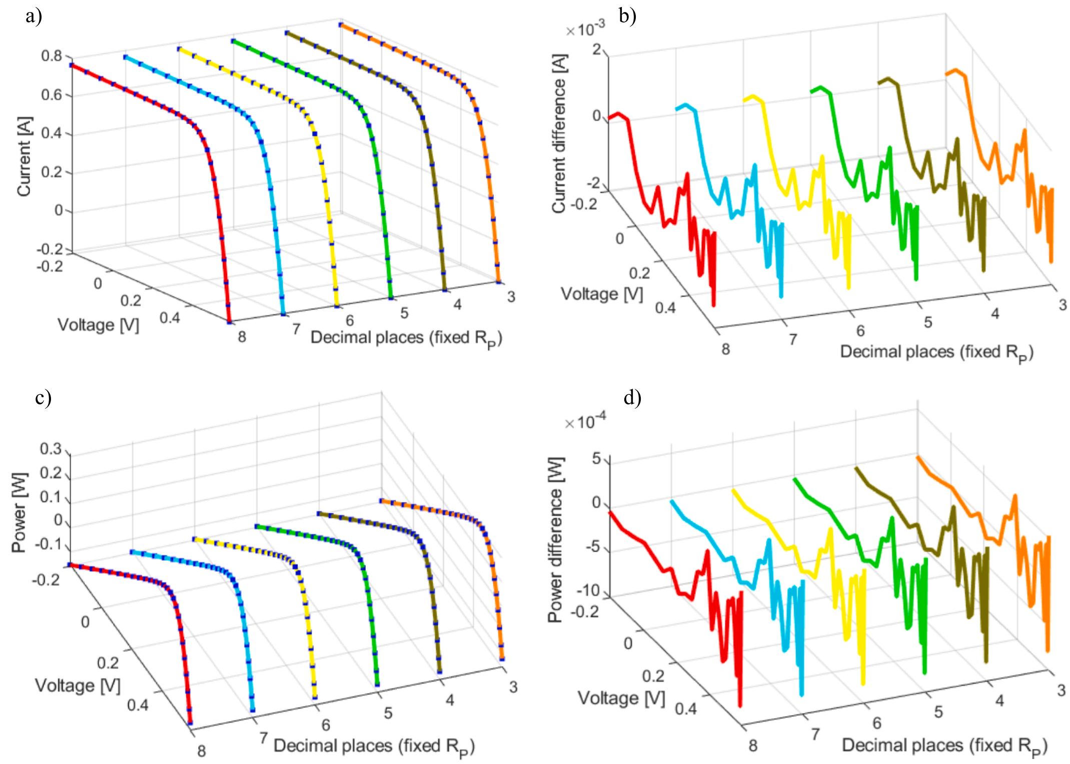


Fig. 13. RTC France solar cell characteristics obtained with fixed R_P at different decimal precisions: a) I–V characteristics b) Current deviation c) P–V characteristics d) Power deviation.

Table 14
The estimated values of parameters for MSX-60 solar module – fixed R_P.

Fixed R_P [Ω]	I_{PV} [A]	I_0 [μA]	n	R_S[Ω]	RMSE
894.28671832	3.81096822	0.12074801	1.32121376	0.22787259	0.011651516098035
894.2867183	3.81105203	0.12079551	1.32124063	0.22785861	0.011651474927588
894.286718	3.81085518	0.11900184	1.32010300	0.22824836	0.011651538866357
894.28671	3.81097633	0.11959767	1.32048349	0.22817649	0.011651420998107
894.2867	3.81094554	0.12002318	1.32075035	0.22800396	0.011651319887925
894.286	3.81116823	0.12167813	1.32178342	0.22753718	0.011652160249601

that R_P can be approximated with a smaller number of significant digits without a noticeable loss of model accuracy, which can simplify the practical implementation of the SDM and reduce data-storage requirements.

The additional analysis of parameter combinations shows that interactions between two parameters can either amplify or partially compensate their overall impact on the error, indicating a non-trivial interplay and mutual dependence among the parameters in the SDM. This observation emphasizes the importance of multidimensional analyses when assessing parameter precision, since the effect of one parameter may depend on the simultaneous variation of another.

Overall, the results indicate that an appropriate balance between numerical precision and practical model accuracy can contribute to more efficient and more stable modeling of solar cells and PV modules. This is particularly important in real-system applications, where excessive precision may increase data overhead without producing a practically meaningful improvement in accuracy, whereas insufficient precision may compromise the reliability of simulations.

A key conclusion is that not all parameters are equally sensitive to reduced precision. Therefore, high numerical precision should not be treated as uniformly necessary for all SDM parameters. Instead, the required precision should be selected according to the sensitivity of each parameter and its influence on the I–V characteristic. The additional normalized sensitivity analysis, based on |Sp|, supports this conclusion by enabling comparison of parameter sensitivity on a common dimensionless scale.

In other words, the paper provides practical guidelines for balancing computational precision and actual model accuracy, thereby contributing to more efficient and reliable modeling of solar cells and PV modules. Therefore, these conclusions should be interpreted as practical guidelines for the parameter estimation of solar cells/modules.

However, although this study provides a detailed examination of how numerical precision and parameter truncation affect SDM parameter representation and model accuracy, several inherent limitations arise from the scope and framework of the investigation:

- Model-level scope: The analysis is intentionally restricted to the classical five-parameter single-diode model, which is widely used as a reference model for PV characterization. Consequently, the findings relate specifically to the behavior of this model structure and its associated numerical properties.
- Precision and uncertainty scope: The study is primarily focused on the impact of numerical truncation, percentage reduction of parameter values, and reduction in the number of significant digits on SDM parameter representation and model accuracy. Current-measurement noise is considered through a Monte Carlo robustness check; however, a complete uncertainty propagation, including

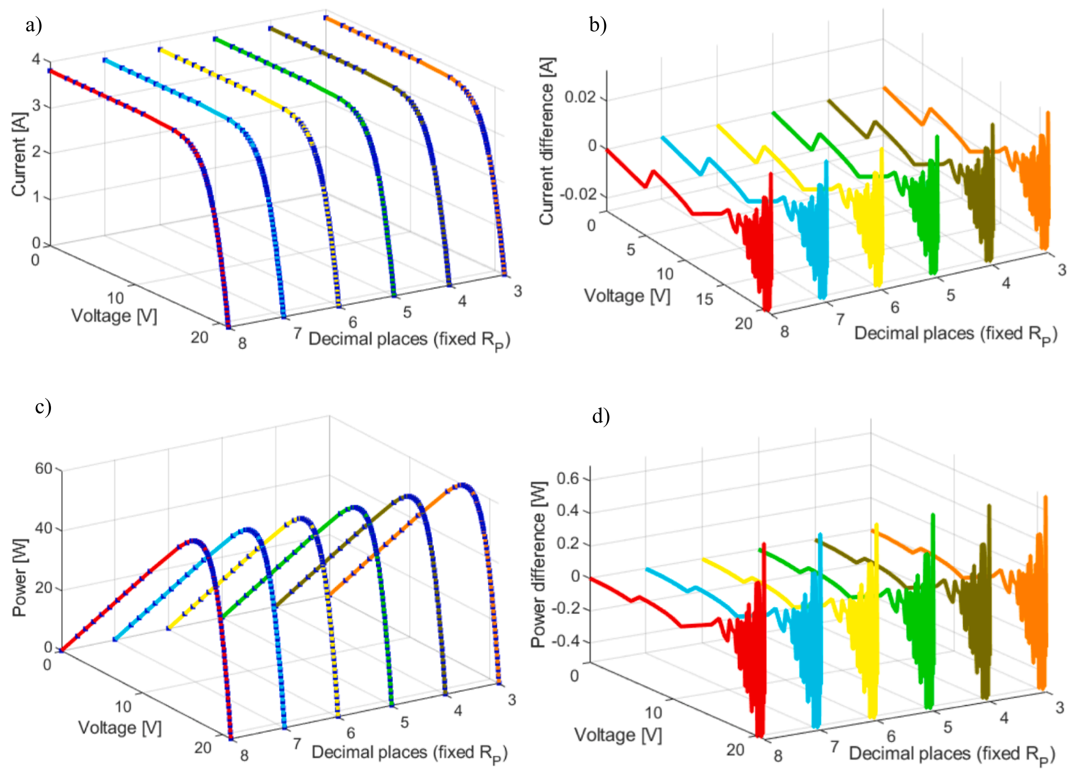


Fig. 14. MSX-60 solar module characteristics obtained with fixed R_P at different decimal precisions: a) I–V characteristics b) Current deviation c) P–V characteristics d) Power deviation.

voltage uncertainty, operating-condition fluctuations, and long-term material effects, is beyond the scope of this work.

- Static I–V characteristic scope: The study is based on the static I–V characteristic, which enables full control over the estimation procedure and numerical precision. This approach provides a consistent framework for evaluating the impact of truncation without introducing additional model dynamics that are beyond the scope of this work.
- Optimization-algorithm scope: The study does not analyze the numerical stability or convergence properties of the optimization algorithms themselves. These algorithms are used solely as tools to obtain parameter estimates, while the focus of the work is on how the numerical precision of the obtained parameters influences model accuracy.
- Dynamic operating conditions: The present study is restricted to static I–V characteristics under STC. The impact of parameter truncation and numerical precision under varying temperature, irradiance, or partial shading conditions has not been investigated and represents a promising direction for future research.
- Transferability of precision thresholds: Although the normalized sensitivity coefficient |Sp| provides a dimensionless basis for comparing parameter sensitivity across the analyzed cells and modules, the exact decimal-place precision thresholds obtained in this study should not be interpreted as universal fixed limits. These thresholds depend on the considered dataset, parameter values, measurement uncertainty, and the operating range of the I–V characteristic. Therefore, they should be regarded as practical guidelines for the analyzed benchmark cases, while their extension to other PV technologies and operating conditions, especially beyond STC, requires additional validation.

7. Conclusion

In this paper, the impact of parameter truncation on the accuracy of solar cell and module modelling within the single-diode model (SDM) is comprehensively analyzed. The study considers both decimal-place truncation and controlled percentage reductions of SDM parameters, allowing the influence of numerical precision on model accuracy to be evaluated from complementary perspectives. The obtained results show that most SDM parameters, including n, I_{PV}, I_0, and R_S, exhibit a certain degree of sensitivity to a reduction in the number of significant digits (parameter truncation), with accuracy degradation occurring only beyond a certain precision threshold. Although the intensity of this sensitivity varies among the parameters, they share the common feature that insufficient numerical precision (due to truncation) can lead to a measurable increase in the modelling error, which highlights the need for their careful estimation and appropriate control of numerical precision.

In contrast, the shunt resistance R_P consistently exhibits extremely low sensitivity to changes in numerical precision (parameter truncation). Even when it is rounded to only a few decimal places, there is virtually no effect on the RMSE value, the shape of the I–V curve, or the position of the maximum power point. Additional analyses in which R_P is fixed to prescribed values confirm that re-estimation of the remaining parameters remains stable and that the overall accuracy of the model is not degraded. This indicates that high numerical precision of R_P is not necessary and that a lower-precision approximation can be used without compromising the physical fidelity of the model. For the analyzed cells and modules and within the investigated precision range, these results indicate that high numerical precision of R_P is generally not required and that a lower-precision approximation can be employed without compromising the physical fidelity of the model.

The performed two-parameter reduction analysis demonstrates that the influence of numerical precision loss in the SDM cannot be fully understood by considering each parameter independently. The obtained results reveal that simultaneous parameter reductions often produce interaction effects that are either compensating or cumulative, depending on the physical role of the parameters and their position within the nonlinear Lambert W formulation. Most parameter pairs exhibited compensating behavior, indicating that the combined RMSE increase was smaller than the sum of the corresponding individual RMSE increases. This finding confirms the existence of significant parameter interdependence within the SDM and highlights the presence of partial error-cancellation mechanisms. On the other hand, several parameter pairs, particularly those involving the diode ideality factor and shunt resistance, showed cumulative behavior, demonstrating that joint parameter reduction may also amplify modelling errors beyond the additive expectation. The results further show that the interaction type may depend on the reduction level, with some parameter pairs transitioning from compensating to cumulative behavior as the reduction becomes stronger. These findings demonstrate that parameter sensitivity within the SDM is not solely an individual property of each parameter, but also depends on nonlinear interactions among model parameters. Overall, the analysis provides a deeper understanding of SDM parameter interactions and confirms that preserving numerical precision is especially important for parameter combinations that strongly influence the nonlinear diode-related terms of the model.

Furthermore, the Monte Carlo current-noise analysis showed that the practical significance of parameter reduction depends not only on the resulting RMSE increase but also on the uncertainty associated with current measurements. For several reduction scenarios, the induced RMSE variation remained comparable to or below the estimated measurement uncertainty, indicating that certain reductions in numerical precision may have limited practical impact despite being detectable from a purely numerical perspective.

In general, the results indicate that high numerical precision for all parameters does not need to be treated as a universal requirement. Instead, a differentiated approach is recommended, in which high precision is reserved for parameters with a strong influence on model accuracy, whereas parameters with lower sensitivity can be represented with reduced numerical precision. In this way, a better balance is achieved between computational efficiency and modelling accuracy, which is of practical importance in simulations, optimization procedures, and implementations in real PV systems.

Moreover, it should be noted that this study is confined to static I–V characteristics under STC. In future work, the influence of parameter truncation and numerical precision under varying operating conditions—such as temperature fluctuations, irradiance variations, and partial shading—should be systematically investigated to assess how dynamic conditions affect the required precision of parameters.

The paper contributes to a quantitative understanding of parameter sensitivity, numerical robustness, and parameter interaction effects within the SDM, while providing practical guidelines for selecting an appropriate numerical precision level without unnecessarily increasing computational complexity. Such an approach enables more efficient, stable, and reliable modelling of solar cells and modules, which is relevant both for academic research and for engineering applications that rely on accurate PV modelling.

8. Statement

During the preparation of this work, the author(s) used ChatGPT in order to edit and improve the English language for clarity and grammar. After using this tool, the author(s) reviewed and edited the content as needed and took full responsibility for the content of the published article.

CRediT authorship contribution statement

Martin Ćalasan: Writing – review & editing, Writing – original draft, Visualization, Validation, Investigation, Formal analysis, Conceptualization. **Snežana Vujošević:** Writing – original draft, Visualization, Validation, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Contains the tables corresponding to the RTC France solar cell

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Table A1
Measured current–voltage (I–V) data for RTC France solar cell under STC conditions.

Voltage [V]	Current [A]	Voltage [V]	Current [A]
−0.2057	0.7640	0.4137	0.7280
−0.1291	0.7620	0.4373	0.7065
−0.0588	0.7605	0.4590	0.6755
0.0057	0.7605	0.4784	0.6320
0.0646	0.7600	0.4960	0.5730
0.1185	0.7590	0.5119	0.4990
0.1678	0.7570	0.5265	0.4130
0.2132	0.7570	0.5398	0.3165
0.2545	0.7555	0.5521	0.2120
0.2924	0.7540	0.5633	0.1035
0.3269	0.7505	0.5736	−0.0100
0.3585	0.7465	0.5833	−0.1230
0.3873	0.7385	0.5900	−0.2100

Table A2
RMSE values for the RTC France solar cell using the WaOA parameter set under decimal-place truncation ($\times 10^{-4}$).

	Decimal places					
	8	7	6	5	4	3
I_{PV}[A]	7.7300626905	7.7300627139	7.7300675148	7.7303896856	7.7698189484	10.4497981821
I_0[μA]	7.7300626905	7.7300626958	7.7300626958	7.7301138792	7.7521752817	9.0808222069
n	7.7300626905	7.7300626932	7.7300735978	7.7317618353	7.8630964480	9.6064228396
R_S[Ω]	7.7300626905	7.7300627186	7.7300729761	7.7306135923	7.7551749999	10.6193759006
R_P[Ω]	7.7300626905	7.7300626905	7.7300626905	7.7300626905	7.7300626920	7.7300627397

Table A3
RMSE values for the RTC France solar cell using the WaOA parameter set under significant-digit truncation ($\times 10^{-4}$).

	Significant digits					
	full precision	7	6	5	4	3
I_{PV}[A]	7.7300626905	7.7300735476	7.7300840662	7.7304507050	7.7703852574	10.4535071308
I_0[μA]	7.7300626905	7.7300735476	7.7300735476	7.7300770736	7.7511880961	9.07389221262
n	7.7300626905	7.7300627124	7.7300735476	7.7317609761	7.8630887680	9.60639773234
R_S[Ω]	7.7300626905	7.7300627257	7.7300727468	7.7306117871	7.7551627199	10.6192704898
R_P[Ω]	7.7300626905	7.7300627108	7.7300627156	7.7300627855	7.7300716682	7.73079835941

Table A4
Normalized decimal-place truncation results for the RTC France solar cell using the WaOA parameter set.

Parameter	Decimal places	p_0	p_{trunc}	$\Delta p/\ p_0$[%]	ΔRMSE/ $RMSE_0$[%]	S_p	$\|S_p\|$
I_{PV}	**7**	0.76078796	0.7607879	−7.88656E-06	3.02421E-07	−0.038346332	0.038346332
	6	0.76078796	0.760787	−0.000126185	6.24096E-05	−0.49458814	0.49458814
	5	0.76078796	0.76078	−0.001046284	0.004230174	−4.043047447	4.043047447
	4	0.76078796	0.7607	−0.011561697	0.514307057	−44.48369902	44.48369902
	3	0.76078796	0.760	−0.103571565	35.18387367	−339.7059175	339.7059175
I_0	**7**	0.31068404	0.310684	−1.28748E-05	6.79301E-08	−0.005276201	0.005276201
	6	0.31068404	0.310684	−1.28748E-05	6.79301E-08	−0.005276201	0.005276201
	5	0.31068404	0.31068	−0.001300356	0.000662203	−0.509247059	0.509247059
	4	0.31068404	0.3106	−0.027049989	0.286059662	−10.57522268	10.57522268
	3	0.31068404	0.310	−0.220172237	17.47410817	−79.36562953	79.36562953
n	**7**	1.47726761	1.4772676	−6.76925E-07	3.54331E-08	−0.052344143	0.052344143
	6	1.47726761	1.477267	−4.12925E-05	0.000141103	−3.417158535	3.417158535
	5	1.47726761	1.47726	−0.00051514	0.021980997	−42.66992718	42.66992718
	4	1.47726761	1.4772	−0.004576693	1.720991961	−376.0339714	376.0339714
	3	1.47726761	1.477	−0.018115201	24.27354375	−1339.95441	1339.95441
R_S	**7**	0.03654695	0.0365469	−0.00013681	3.62987E-07	−0.002653212	0.002653212
	6	0.03654695	0.036546	−0.002599396	0.000133059	−0.051188509	0.051188509
	5	0.03654695	0.03654	−0.019016635	0.007126744	−0.374763701	0.374763701
	4	0.03654695	0.0365	−0.128464892	0.324865534	−2.528827358	2.528827358
	3	0.03654695	0.036	−1.496568113	37.37761679	−24.97555338	24.97555338
R_P	**7**	52.88969619	52.8896961	−1.70165E-07	−1.40258E-12	8.24244E-06	8.24244E-06
	6	52.88969619	52.889696	−3.59238E-07	2.09686E-11	−5.83695E-05	5.83695E-05
	5	52.88969619	52.88969	−1.17036E-05	6.54401E-10	−5.59145E-05	5.59145E-05
	4	52.88969619	52.8896	−0.000181869	2.02757E-08	−0.000111485	0.000111485
	3	52.88969619	52.889	−0.001316306	6.36615E-07	−0.000483638	0.000483638

Table A5
Controlled percentage perturbation results for the RTC France solar cell using the WaOA parameter set.

Parameter	Perturbation [%]	p_0	p_{new}	$\Delta p/\ p_0$[%]	ΔRMSE/ $RMSE_0$[%]	S_p	$\|S_p\|$
I_{PV}	−5	0.76078796	0.722748562	−5	4296.29254	−859.258508	859.258508
	−1	0.76078796	0.75318008	−1	784.0885555	−784.0885555	784.0885555
	−0.1	0.76078796	0.760027172	−0.1	33.09330042	−330.9330042	330.9330042
	0.1	0.76078796	0.761548748	0.1	33.09006956	330.9006956	330.9006956
	1	0.76078796	0.76839584	1	783.7797486	783.7797486	783.7797486
	5	0.76078796	0.798827358	5	4288.567922	857.7135845	857.7135845
I_0	−5	0.31068404	0.295149838	−5	1324.384094	−264.8768188	264.8768188
	−1	0.31068404	0.3075772	−1	197.9407741	−197.9407741	197.9407741
	−0.1	0.31068404	0.310373356	−0.1	3.842943661	−38.42943661	38.42943661

(*continued on next page*)

Table A5 (*continued*)

Parameter	Perturbation [%]	p_0	p_{new}	$\Delta p/ p_0$[%]	ΔRMSE/ RMSE$_0$[%]	S_p	$\|S_p\|$
	0.1	0.31068404	0.310994724	0.1	3.838291335	38.38291335	38.38291335
	1	0.31068404	0.31379088	1	196.3306709	196.3306709	196.3306709
	5	0.31068404	0.326218242	5	1281.725087	256.3450173	256.3450173
n	−5	1.47726761	1.40340423	−5	23785.54245	−4757.108489	4757.108489
	−1	1.47726761	1.462494934	−1	4109.612236	−4109.612236	4109.612236
	−0.1	1.47726761	1.475790342	−0.1	320.4844213	−3204.844213	3204.844213
	0.1	1.47726761	1.478744878	0.1	317.8264576	3178.264576	3178.264576
	1	1.47726761	1.492040286	1	3836.059516	3836.059516	3836.059516
	5	1.47726761	1.551130991	5	16993.90895	3398.781789	3398.781789
R_S	−5	0.03654695	0.034719603	−5	231.9975124	−46.39950247	46.39950247
	−1	0.03654695	0.036181481	−1	18.13089562	−18.13089562	18.13089562
	−0.1	0.03654695	0.036510403	−0.1	0.196955891	−1.969558914	1.969558914
	0.1	0.03654695	0.036583497	0.1	0.196834007	1.968340074	1.968340074
	1	0.03654695	0.03691242	1	18.02024078	18.02024078	18.02024078
	5	0.03654695	0.038374298	5	227.0339976	45.40679952	45.40679952
R_P	−5	52.88969619	50.24521138	−5	8.706552046	−1.741310409	1.741310409
	−1	52.88969619	52.36079923	−1	0.334135255	−0.334135255	0.334135255
	−0.1	52.88969619	52.83680649	−0.1	0.003291533	−0.032915334	0.032915334
	0.1	52.88969619	52.94258589	0.1	0.003268034	0.032680345	0.032680345
	1	52.88969619	53.41859315	1	0.320953819	0.320953819	0.320953819
	5	52.88969619	55.534181	5	7.179338839	1.435867768	1.435867768

Appendix B. Contains the tables corresponding to the MSX-60 solar module

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Table B1
Measured current–voltage (I–V) data for MSX-60 solar module under STC conditions.

Voltage [V]	Current [A]	Voltage [V]	Current [A]	Voltage [V]	Current [A]
0.0000	3.8100	2.0000	3.8100	3.0000	3.8100
4.0000	3.8100	5.0000	3.8100	6.0000	3.8000
7.0000	3.8000	8.0000	3.8000	9.0000	3.8000
10.0000	3.8000	11.0200	3.8000	14.6900	3.7500
15.6200	3.7000	16.2400	3.6500	16.6300	3.6000
16.8800	3.5500	17.1400	3.5000	17.3500	3.4500
17.5300	3.4000	17.7100	3.3500	17.8200	3.3000
17.9600	3.2500	18.1100	3.2000	18.2200	3.1500
18.2900	3.1000	18.4000	3.0500	18.5000	3.0000
18.5800	2.9500	18.6500	2.9000	18.7600	2.8500
18.8300	2.8000	18.9000	2.7500	18.9400	2.7000
19.0400	2.6500	19.0800	2.6000	19.1500	2.5500
19.1900	2.5000	19.3300	2.4000	19.4400	2.3000
19.5500	2.2000	19.6200	2.1000	19.7300	2.0000
19.8000	1.9000	19.9100	1.8000	19.9800	1.7000
20.0500	1.6000	20.1200	1.5000	20.2300	1.4000
20.2700	1.3000	20.3400	1.2000	20.4100	1.1000
20.4800	1.0000	20.5600	0.9000	20.6300	0.8000
20.7000	0.7000	20.7400	0.6000	20.8100	0.5000
20.8800	0.4000	20.9200	0.3000	20.9900	0.2000
21.0200	0.1000	21.1000	0.0000		

Table B2
RMSE values for the MSX-60 solar module using the AWaOA parameter set under decimal-place truncation ($\times 10^{-2}$).

Parameter	Decimal places				
	8	7	6	5	4
I_{PV}[A]	1.1651382745	1.1651382835	1.1651383872	1.1651387681	1.1651630284
I_0[μA]	1.1651382745	1.1651382542	1.1651373475	1.1651362654	1.1672309129
n	1.1651382745	1.1651384963	1.1651412416	1.1652040799	1.1745626932
R_S[Ω]	1.1651382745	1.1651382643	1.1651382440	1.1651374540	1.1651621662
R_P[Ω]	1.1651382745	1.1651382745	1.1651382745	1.1651382743	1.1651382741

Table B3
RMSE values for the MSX-60 solar module using the AWaOA parameter set under significant-digit truncation ($\times 10^{-2}$).

Parameter	Significant digits					
	8	7	6	5	4	3
I_{PV}[A]	1.1651382745	1.1651413614	1.1651419054	1.1651699704	1.1681533166	1.1681533166
I_0[μA]	1.1651382745	1.1651413614	1.1651399433	1.1651357933	1.1671589892	1.2932210627
n	1.1651382745	1.1651413614	1.1652045019	1.1745676094	1.1745676094	2.0315634455
R_S[Ω]	1.1651382745	1.1651413614	1.1651413294	1.1651399498	1.1651575863	1.1714112982
R_P[Ω]	1.1651382745	1.1651413614	1.1651413459	1.1651412133	1.1651394543	1.1651351323

Table B4
Normalized decimal-place truncation results for the MSX-60 solar module using the AWaOA parameter set.

Parameter	Decimal places	p_0	p_{trunc}	$\Delta p/p_0$[%]	ΔRMSE/ RMSE$_0$[%]	S_p	$\|S_p\|$
I_{PV}	**7**	3.81097398	3.8109739	−2.0992E-06	7.70947E-07	−0.367257417	0.367257417
	6	3.81097398	3.810973	−2.57152E-05	9.67184E-06	−0.376113733	0.376113733
	5	3.81097398	3.81097	−0.000104435	4.23625E-05	−0.405633846	0.405633846
	4	3.81097398	3.8109	−0.001941236	0.002124544	−1.094428465	1.094428465
I_0	**7**	0.12057351	0.1205735	−8.2937E-06	−1.73816E-06	0.209575853	0.209575853
	6	0.12057351	0.120573	−0.000422978	−7.95617E-05	0.188098697	0.188098697
	5	0.12057351	0.12057	−0.002911087	−0.000172435	0.059233903	0.059233903
	4	0.12057351	0.1205	−0.060966957	0.179604296	−2.945928492	2.945928492
n	**7**	1.32109677	1.3210967	−5.29863E-06	1.90388E-05	−3.593158771	3.593158771
	6	1.32109677	1.321096	−5.82849E-05	0.00025466	−4.369221126	4.369221126
	5	1.32109677	1.32109	−0.000512453	0.005647864	−11.02123417	11.02123417
	4	1.32109677	1.321	−0.007324974	0.808866977	−110.425912	110.425912
R_S	**7**	0.22786515	0.2278651	−2.19428E-05	−8.75525E-07	0.039900334	0.039900334
	6	0.22786515	0.227865	−6.58284E-05	−2.6153E-06	0.039729058	0.039729058
	5	0.22786515	0.22786	−0.002260109	−7.04169E-05	0.031156432	0.031156432
	4	0.22786515	0.2278	−0.028591472	0.002050543	−0.071718705	0.071718705
R_P	**7**	894.2867183	894.2867183	−2.23642E-09	−3.7147E-11	0.016610006	0.016610006
	6	894.2867183	894.286718	−3.57827E-08	−6.42754E-10	0.017962702	0.017962702
	5	894.2867183	894.28671	−9.3035E-07	−1.68516E-08	0.018113133	0.018113133
	4	894.2867183	894.2867	−2.04856E-06	−3.71142E-08	0.018117197	0.018117197

Table B5
Controlled percentage perturbation results for the MSX-60 solar module using the AWaOA parameter set.

Parameter	Perturbation [%]	p_0	p_{new}	$\Delta p/p_0$[%]	ΔRMSE/ RMSE$_0$[%]	S_p	$\|S_p\|$
I_{PV}	−5	3.81097398	3.620425281	−5	1275.454451	−255.0908902	255.0908902
	−1	3.81097398	3.77286424	−1	191.7633541	−191.7633541	191.7633541
	−0.1	3.81097398	3.807163006	−0.1	3.717822915	−37.17822915	37.17822915
	0.1	3.81097398	3.814784954	0.1	3.646629892	36.46629892	36.46629892
	1	3.81097398	3.84908372	1	191.3334534	191.3334534	191.3334534
	5	3.81097398	4.001522679	5	1270.449411	254.0898822	254.0898822
I_0	−5	0.12057351	0.114544835	−5	425.1530265	−85.03060529	85.03060529
	−1	0.12057351	0.119367775	−1	42.72826331	−42.72826331	42.72826331
	−0.1	0.12057351	0.120452936	−0.1	0.495957154	−4.959571541	4.959571541
	0.1	0.12057351	0.120694084	0.1	0.537197251	5.371972512	5.371972512
	1	0.12057351	0.121779245	1	42.63860281	42.63860281	42.63860281
	5	0.12057351	0.126602186	5	412.3447201	82.46894402	82.46894402
n	−5	1.32109677	1.255041932	−5	10532.63183	−2106.526367	2106.526367
	−1	1.32109677	1.307885802	−1	1693.715671	−1693.715671	1693.715671
	−0.1	1.32109677	1.319775673	−0.1	99.07692873	−990.7692873	990.7692873
	0.1	1.32109677	1.322417867	0.1	97.37370087	973.7370087	973.7370087
	1	1.32109677	1.334307738	1	1537.451085	1537.451085	1537.451085
	5	1.32109677	1.387151609	5	6666.629316	1333.325863	1333.325863
R_S	−5	0.22786515	0.216471893	−5	72.19372831	−14.43874566	14.43874566
	−1	0.22786515	0.225586499	−1	3.800816687	−3.800816687	3.800816687
	−0.1	0.22786515	0.227637285	−0.1	0.035069261	−0.350692612	0.350692612
	0.1	0.22786515	0.228093015	0.1	0.043050878	0.430508777	0.430508777
	1	0.22786515	0.230143802	1	3.865793269	3.865793269	3.865793269
	5	0.22786515	0.239258408	5	71.51681113	14.30336223	14.30336223
R_P	−5	894.2867183	849.5723824	−5	0.165728098	−0.03314562	0.03314562
	−1	894.2867183	885.3438511	−1	−0.008679713	0.008679713	0.008679713
	−0.1	894.2867183	893.3924316	−0.1	−0.001719127	0.017191268	0.017191268
	0.1	894.2867183	895.181005	0.1	0.001904072	0.019040723	0.019040723
	1	894.2867183	903.2295855	1	0.027179051	0.027179051	0.027179051
	5	894.2867183	939.0010542	5	0.299659076	0.059931815	0.059931815

Appendix C. Contains the tables corresponding to the Photowatt-PWP 201 solar module

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Table C1
Measured current–voltage (I–V) data for Photowatt-PWP 201 solar module under STC conditions.

Voltage [V]	Current [A]	Voltage [V]	Current [A]
−1.9426	1.0345	13.1231	0.8725
0.1248	1.0315	13.6983	0.8075
1.8093	1.0300	14.2221	0.7265
3.3511	1.0260	14.6995	0.6345
4.7622	1.0220	15.1346	0.5345
6.0538	1.0180	15.5311	0.4275
7.2364	1.0155	15.8929	0.3185
8.3189	1.0140	16.2229	0.2085
9.3097	1.0100	16.5241	0.1010
10.2163	1.0035	16.7987	−0.0080
11.0449	0.9880	17.0499	−0.1110
11.8018	0.9630	17.2793	−0.2090
12.4929	0.9255	17.4885	−0.3030

Table C2
RMSE values for the Photowatt-PWP 201 solar module using the CGTO parameter set under decimal-place truncation ($\times 10^{-3}$).

Parameter	Decimal places					
	8	7	6	5	4	3
I_{PV}[A]	2.0399922732	2.0399922746	2.0399923322	2.0400020358	2.0405541973	2.0615463210
I_0[μA]	2.0399922732	2.0399922732	2.0399922749	2.0399923844	2.0400235663	2.0412044545
n	2.0399922732	2.0399922747	2.0399928309	2.0400775114	2.0413995083	2.7151907635
R_S[Ω]	2.0399922732	2.0399922732	2.0399922733	2.0399923428	2.0399951799	2.0403810717
R_P[Ω]	2.0399922732	2.0399922732	2.0399922732	2.0399922732	2.0399922732	2.0399922732

Table C3
RMSE values for the Photowatt-PWP 201 solar module using the CGTO parameter set under significant-digit truncation ($\times 10^{-3}$).

Parameter	Significant digits					
	full precision	7	6	5	4	3
I_{PV}[A]	2.0399922732	2.0399929904	2.0400045829	2.0405702326	2.0616422925	2.8297332348
I_0[μA]	2.0399922732	2.0399929904	2.0399925947	2.0400146830	2.0411450837	2.1831951685
n	2.0399922732	2.0399929904	2.0400789128	2.0414050459	2.7152882081	19.268404085
R_S[Ω]	2.0399922732	2.0399929904	2.0399929360	2.0399950649	2.0403721095	2.0403721095
R_P[Ω]	2.0399922732	2.0399929904	2.0399930006	2.0399930238	2.0399932863	2.0400038534

Table C4
Normalized decimal-place truncation results for the Photowatt-PWP 201 solar module using the CGTO parameter set.

Parameter	Decimal places	p_0	p_{trunc}	$\Delta p/\ p_0$[%]	ΔRMSE/ $RMSE_0$[%]	S_p	$\lvert S_p\rvert$
I_{PV}	**7**	1.03235759	1.0323575	−8.71791E-06	6.78994E-08	−0.007788493	0.007788493
	6	1.03235759	1.032357	−5.71507E-05	2.89546E-06	−0.050663563	0.050663563
	5	1.03235759	1.03235	−0.00073521	0.000478561	−0.650916481	0.650916481
	4	1.03235759	1.0323	−0.005578493	0.027545404	−4.937785537	4.937785537
	3	1.03235759	1.032	−0.034638192	1.056574972	−30.50317939	30.50317939
I_0	**7**	2.49659566	2.4965956	−2.40327E-06	1.53251E-09	−0.000637678	0.000637678
	6	2.49659566	2.496595	−2.6436E-05	8.31559E-08	−0.003145557	0.003145557
	5	2.49659566	2.49659	−0.000226709	5.45248E-06	−0.024050616	0.024050616
	4	2.49659566	2.4965	−0.003831618	0.001533981	−0.400348144	0.400348144
	3	2.49659566	2.496	−0.02385889	0.059420879	−2.490513177	2.490513177
n	**7**	1.31662653	1.3166265	−2.27855E-06	7.69238E-08	−0.033759962	0.033759962
	6	1.31662653	1.316626	−4.02544E-05	2.73415E-05	−0.679218018	0.679218018
	5	1.31662653	1.31662	−0.000495964	0.00417836	−8.424716869	8.424716869
	4	1.31662653	1.3166	−0.002014998	0.068982374	−34.23446028	34.23446028
	3	1.31662653	1.316	−0.047586008	33.09809057	−695.5424983	695.5424983
R_S	**7**	1.24054732	1.2405473	−1.61219E-06	4.51753E-11	−2.8021E-05	2.8021E-05

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Table C4 (*continued*)

Parameter	Decimal places	p_0	p_{trunc}	$\Delta p/\ p_0$[%]	ΔRMSE/ $RMSE_0$[%]	S_p	$\|S_p\|$
	6	1.24054732	1.240547	−2.57951E-05	6.75707E-09	−0.000261952	0.000261952
	5	1.24054732	1.24054	−0.000590062	3.4143E-06	−0.005786346	0.005786346
	4	1.24054732	1.2405	−0.003814445	0.000142487	−0.037354499	0.037354499
	3	1.24054732	1.240	−0.044119236	0.019058825	−0.431984476	0.431984476
R_P	**7**	748.3229419	748.3229418	−9.35426E-09	4.08172E-12	−0.000436349	0.000436349
	6	748.3229419	748.322941	−1.1626E-07	−3.72032E-12	3.2E-05	3.2E-05
	5	748.3229419	748.32294	−2.49892E-07	1.89205E-12	−7.57145E-06	7.57145E-06
	4	748.3229419	748.3229	−5.59518E-06	−2.14715E-11	3.83751E-06	3.83751E-06
	3	748.3229419	748.322	−0.000125864	2.37703E-09	−1.88857E-05	1.88857E-05

Table C5
Controlled percentage perturbation results for the Photowatt-PWP 201 solar module using the CGTO parameter set.

Parameter	Perturbation [%]	p_0	p_{new}	$\Delta p/\ p_0$[%]	ΔRMSE/ $RMSE_0$[%]	S_p	$\|S_p\|$
I_{PV}	−5	1.03235759	0.980739711	−5	2009.692277	−401.9384554	401.9384554
	−1	1.03235759	1.022034014	−1	332.6278614	−332.6278614	332.6278614
	−0.1	1.03235759	1.031325232	−0.1	8.492503311	−84.92503311	84.92503311
	0.1	1.03235759	1.033389948	0.1	8.491418362	84.91418362	84.91418362
	1	1.03235759	1.042681166	1	332.3589365	332.3589365	332.3589365
	5	1.03235759	1.08397547	5	2002.790598	400.5581195	400.5581195
I_0	−5	2.49659566	2.371765877	−5	642.070993	−128.4141986	128.4141986
	−1	2.49659566	2.471629703	−1	76.13209736	−76.13209736	76.13209736
	−0.1	2.49659566	2.494099064	−0.1	1.039190719	−10.39190719	10.39190719
	0.1	2.49659566	2.499092256	0.1	1.037680114	10.37680114	10.37680114
	1	2.49659566	2.521561617	1	75.30696952	75.30696952	75.30696952
	5	2.49659566	2.621425443	5	617.2154759	123.4430952	123.4430952
n	−5	1.31662653	1.25079520	−5	10243.60186	−2048.720371	2048.720371
	−1	1.31662653	1.30346026	−1	1790.96636	−1790.96636	1790.96636
	−0.1	1.31662653	1.31530990	−0.1	110.135919	−1101.35919	1101.35919
	0.1	1.31662653	1.31794316	0.1	109.3546737	1093.546737	1093.546737
	1	1.31662653	1.32979280	1	1702.184943	1702.184943	1702.184943
	5	1.31662653	1.38245786	5	8031.377541	1606.275508	1606.275508
R_S	−5	1.24054732	1.178519954	−5	144.8684047	−28.97368094	28.97368094
	−1	1.24054732	1.228141847	−1	9.388943846	−9.388943846	9.388943846
	−0.1	1.24054732	1.239306773	−0.1	0.097895084	−0.978950844	0.978950844
	0.1	1.24054732	1.241787867	0.1	0.097814912	0.978149121	0.978149121
	1	1.24054732	1.252952793	1	9.317264416	9.317264416	9.317264416
	5	1.24054732	1.302574686	5	140.8306304	28.16612608	28.16612608
R_P	−5	748.3229419	710.9067948	−5	5.115167129	−1.023033426	1.023033426
	−1	748.3229419	740.8397125	−1	0.19303617	−0.19303617	0.19303617
	−0.1	748.3229419	747.5746189	−0.1	0.001897105	−0.018971054	0.018971054
	0.1	748.3229419	749.0712648	0.1	0.001890519	0.018905189	0.018905189
	1	748.3229419	755.8061713	1	0.185484005	0.185484005	0.185484005
	5	748.3229419	785.739089	5	4.205984937	0.841196987	0.841196987

Appendix D. Contains the tables corresponding to the CdTe75669 solar module

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Table D1
RMSE values for the CdTe75669 solar module using the ERWCA parameter set under decimal-place truncation ($\times 10^{-4}$).

Parameter	Decimal places					
	8	7	6	5	4	3
I_{PV}[A]	9.5068649100	9.5068649134	9.5068686402	9.5068809887	9.5185067190	10.0276311769
I_0[μA]	9.5068649100	9.5068649100	9.5068649098	9.5068649155	9.5068764500	9.5082049811
n	9.5068649100	9.5068649229	9.5068667009	9.5074109419	9.5660647033	12.7808701630
R_S[Ω]	9.5068649100	9.5068649100	9.5068649099	9.5068649202	9.5068699134	9.5068928570
R_P[Ω]	9.5068649100	9.5068649100	9.5068649100	9.5068649100	9.5068649101	9.5068649110

Table D2
RMSE values for the CdTe75669 solar module using the ERWCA parameter set under significant-digit truncation ($\times 10^{-4}$).

Parameter	Significant digits					
	full precision	7	6	5	4	3
I_{PV}[A]	9.5068649100	9.5068686402	9.5068809887	9.5185067190	10.0276311769	23.3401139770
I_0[μA]	9.5068649100	9.5068649098	9.5068649155	9.5068764500	9.5082049811	9.5082049811
n	9.5068649100	9.5068667009	9.5074109419	9.5660647033	12.7808701630	95.4481764640
R_S[Ω]	9.5068649100	9.5068649099	9.5068649202	9.5068699134	9.5068928570	9.5112570060
R_P[Ω]	9.5068649100	9.5068649110	9.5068649748	9.5068652922	9.5069975034	9.5315402669

Table D3
Normalized decimal-place truncation results for the CdTe75669 solar module using the ERWCA parameter set.

Parameter	Decimal places	p_0	p_{trunc}	$\Delta p/ p_0$[%]	ΔRMSE/ $RMSE_0$[%]	S_p	$\|S_p\|$
I_{PV}	**7**	1.16235192	1.1623519	−1.72065E-06	3.5079E-08	−0.020387068	0.020387068
	6	1.16235192	1.162351	−7.91499E-05	3.92367E-05	−0.495727173	0.495727173
	5	1.16235192	1.16235	−0.000165182	0.000169127	−1.023880631	1.023880631
	4	1.16235192	1.1623	−0.004466806	0.122456867	−27.41486418	27.41486418
	3	1.16235192	1.162	−0.030276545	5.477791805	−180.925262	180.925262
I_0	**7**	6.20055140	6.2005514	0	0	not defined*	not defined*
	6	6.20055140	6.200551	−6.45104E-06	−2.02432E-09	0.000313797	0.000313797
	5	6.20055140	6.20055	−2.25786E-05	5.78724E-08	−0.002563149	0.002563149
	4	6.20055140	6.2005	−0.000828959	0.000121386	−0.146431892	0.146431892
	3	6.20055140	6.200	−0.008892758	0.014095826	−1.585090512	1.585090512
n	**7**	1.94879954	1.9487995	−2.05255E-06	1.35047E-07	−0.06579469	0.06579469
	6	1.94879954	1.948799	−2.77094E-05	1.88382E-05	−0.679848759	0.679848759
	5	1.94879954	1.94879	−0.000489532	0.005743554	−11.73274057	11.73274057
	4	1.94879954	1.9487	−0.00510776	0.622705738	−121.9136684	121.9136684
	3	1.94879954	1.948	−0.041027309	34.43832729	−839.4001099	839.4001099
R_S	**7**	7.02217388	7.0221738	−1.13925E-06	−7.65316E-10	0.000671773	0.000671773
	6	7.02217388	7.022173	−1.25317E-05	−1.56029E-09	0.000124508	0.000124508
	5	7.02217388	7.02217	−5.52535E-05	1.06983E-07	−0.001936215	0.001936215
	4	7.02217388	7.0221	−0.001052096	5.2629E-05	−0.050023055	0.050023055
	3	7.02217388	7.022	−0.002476156	0.000293966	−0.118718674	0.118718674
R_P	**7**	1624.316481	1624.316481	−1.84693E-09	5.62238E-12	−0.003044182	0.003044182
	6	1624.316481	1624.316481	−2.03162E-08	7.90326E-12	−0.000389012	0.000389012
	5	1624.316481	1624.31648	−8.18806E-08	2.11438E-11	−0.000258227	0.000258227
	4	1624.316481	1624.3164	−5.00703E-06	1.33679E-09	−0.000266983	0.000266983
	3	1624.316481	1624.316	−2.96328E-05	1.05824E-08	−0.000357117	0.000357117

* In this case, decimal-place truncation does not change the parameter value, i.e., $\Delta p/p_0 = 0$, so the normalized sensitivity coefficient |Sp| cannot be calculated.

Table D4
Controlled percentage perturbation results for the CdTe75669 solar module using the ERWCA parameter set.

Parameter	Perturbation [%]	p_0	p_{new}	$\Delta p/ p_0$[%]	ΔRMSE/ $RMSE_0$[%]	S_p	$\|S_p\|$
I_{PV}	−5	1.16235192	1.104234324	−5	5448.028152	−1089.60563	1089.60563
	−1	1.16235192	1.150728401	−1	1012.838708	−1012.838708	1012.838708
	−0.1	1.16235192	1.161189568	−0.1	49.26092225	−492.6092225	492.6092225
	0.1	1.16235192	1.163514272	0.1	49.25554285	492.5554285	492.5554285
	1	1.16235192	1.173975439	1	1012.291885	1012.291885	1012.291885
	5	1.16235192	1.220469516	5	5434.346958	1086.869392	1086.869392
I_0	−5	6.2005514	5.89052383	−5	865.4471163	−173.0894233	173.0894233
	−1	6.2005514	6.138545886	−1	114.2802126	−114.2802126	114.2802126
	−0.1	6.2005514	6.194350849	−0.1	1.769481526	−17.69481526	17.69481526
	0.1	6.2005514	6.206751951	0.1	1.767486284	17.67486284	17.67486284
	1	6.2005514	6.262556914	1	113.1944367	113.1944367	113.1944367
	5	6.2005514	6.51057897	5	834.8463022	166.9692604	166.9692604
n	−5	1.94879954	1.851359563	−5	12237.07514	−2447.415027	2447.415027
	−1	1.94879954	1.929311545	−1	2144.740052	−2144.740052	2144.740052
	−0.1	1.94879954	1.94685074	−0.1	141.0499718	−1410.499718	1410.499718
	0.1	1.94879954	1.95074834	0.1	140.0604619	1400.604619	1400.604619
	1	1.94879954	1.968287535	1	2036.221533	2036.221533	2036.221533
	5	1.94879954	2.046239517	5	9530.155228	1906.031046	1906.031046
R_S	−5	7.02217388	6.671065186	−5	402.943618	−80.58872359	80.58872359
	−1	7.02217388	6.951952141	−1	40.22015718	−40.22015718	40.22015718
	−0.1	7.02217388	7.015151706	−0.1	0.481230437	−4.812304369	4.812304369
	0.1	7.02217388	7.029196054	0.1	0.481236231	4.812362307	4.812362307
	1	7.02217388	7.092395619	1	40.12130716	40.12130716	40.12130716
	5	7.02217388	7.373282574	5	399.4530307	79.89060614	79.89060614
R_P	−5	1624.316481	1543.100657	−5	73.98963177	−14.79792635	14.79792635

(*continued on next page*)

Table D4 (*continued*)

Parameter	Perturbation [%]	p_0	p_{new}	$\Delta p/p_0$[%]	ΔRMSE/ $RMSE_0$[%]	S_p	$\|S_p\|$
	−1	1624.316481	1608.073317	−1	3.666962177	−3.666962177	3.666962177
	−0.1	1624.316481	1622.692165	−0.1	0.036688952	−0.366889517	0.366889517
	0.1	1624.316481	1625.940798	0.1	0.036493167	0.364931672	0.364931672
	1	1624.316481	1640.559646	1	3.525381026	3.525381026	3.525381026
	5	1624.316481	1705.532305	5	63.09493829	12.61898766	12.61898766

Appendix E. Contains the tables corresponding to the KC200GT solar module

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Table E1
RMSE values for the KC200GT solar module using the ERWCA parameter set under decimal-place truncation ($\times10^{-3}$).

Parameter	Decimal places					
	8	7	6	5	4	3
I_{PV}[A]	5.1031656936	5.1031656935	5.1031656989	5.1031660594	5.1032006868	5.1627897530
I_0[μA]	5.1031656936	5.1031657026	5.1031825184	5.1061017202	5.1675729566	17.6203742432
n	5.1031656936	5.1031657598	5.1031668860	5.1034819897	5.5534413560	11.4061942181
R_S[Ω]	5.1031656936	5.1031656970	5.1031662070	5.1031814051	5.1038214235	5.1210459454
R_P[Ω]	5.1031656936	5.1031656936	5.1031656936	5.1031656936	5.1031656936	5.1031656929

Table E2
RMSE values for the KC200GT solar module using the ERWCA parameter set under significant-digit truncation ($\times10^{-3}$).

Parameter	Significant digits					
	full precision	7	6	5	4	3
I_{PV}[A]	5.1031656936	5.1031657013	5.1031660728	5.1032008096	5.1627947421	5.3573781544
I_0[μA]	5.1031656936	5.1031656936	5.1031657026	5.1031825184	5.1061017202	5.1675729566
n	5.1031656936	5.1031668860	5.1034819897	5.5534413560	11.4061942181	91.2301148098
R_S[Ω]	5.1031656936	5.1031656970	5.1031662070	5.1031814051	5.1038214235	5.1210459454
R_P[Ω]	5.1031656936	5.1031656936	5.1031656929	5.1031657273	5.1031778648	5.1036669316

Table E3
Normalized decimal-place truncation results for the KC200GT solar module using the ERWCA parameter set.

Parameter	Decimal places	p_0	p_{trunc}	$\Delta p/p_0$[%]	ΔRMSE/ $RMSE_0$[%]	S_p	$\|S_p\|$
I_{PV}	**7**	8.21192233	8.2119223	−3.65323E-07	−3.03574E-09	0.008309741	0.008309741
	6	8.21192233	8.211922	−4.01855E-06	1.03315E-07	−0.025709415	0.025709415
	5	8.21192233	8.21192	−2.83734E-05	7.16769E-06	−0.252620266	0.252620266
	4	8.21192233	8.2119	−0.000271922	0.000685715	−2.521738358	2.521738358
	3	8.21192233	8.211	−0.011231597	1.168373965	−104.0256335	104.0256335
I_0	**7**	0.05752541	0.0575254	−1.73836E-05	1.74896E-07	−0.010060943	0.010060943
	6	0.05752541	0.057525	−0.000712729	0.000329694	−0.462579577	0.462579577
	5	0.05752541	0.05752	−0.00940454	0.057533435	−6.117623752	6.117623752
	4	0.05752541	0.0575	−0.044171784	1.26210409	−28.57263096	28.57263096
	3	0.05752541	0.057	−0.913352899	245.2832085	−268.5525044	268.5525044
n	**7**	1.26338213	1.2633821	−2.37458E-06	1.2965E-06	−0.545991036	0.545991036
	6	1.26338213	1.263382	−1.02898E-05	2.33647E-05	−2.270653206	2.270653206
	5	1.26338213	1.26338	−0.000168595	0.006198036	−36.76285556	36.76285556
	4	1.26338213	1.2633	−0.006500804	8.823457622	−1357.287067	1357.287067
	3	1.26338213	1.263	−0.030246589	123.5121276	−4083.506003	4083.506003
R_S	**7**	0.23712363	0.2371236	−1.26516E-05	6.58944E-08	−0.005208374	0.005208374
	6	0.23712363	0.237123	−0.000265684	1.00588E-05	−0.037860147	0.037860147
	5	0.23712363	0.23712	−0.001530847	0.000307877	−0.201115662	0.201115662
	4	0.23712363	0.2371	−0.009965266	0.012849472	−1.289425957	1.289425957
	3	0.23712363	0.237	−0.05213736	0.350375686	−6.72024221	6.72024221
R_P	**7**	373.4745146	373.4745145	−1.60654E-08	−5.4049E-11	0.003364322	0.003364322
	6	373.4745146	373.474514	−1.49943E-07	−6.93969E-11	0.000462821	0.000462821
	5	373.4745146	373.47451	−1.22097E-06	−2.87446E-10	0.000235425	0.000235425
	4	373.4745146	373.4745	−3.89853E-06	−7.69332E-10	0.000197339	0.000197339
	3	373.4745146	373.474	−0.000137776	−1.46415E-08	0.00010627	0.00010627

Table E4
Controlled percentage perturbation results for the KC200GT solar module using the ERWCA parameter set.

Parameter	Perturbation [%]	p_0	p_{new}	$\Delta p/\ p_0$[%]	ΔRMSE/ $RMSE_0$[%]	S_p	$\|S_p\|$
I_{PV}	−5	8.21192233	7.801326214	−5	6739.229917	−1347.845983	1347.845983
	−1	8.21192233	8.129803107	−1	1269.244542	−1269.244542	1269.244542
	−0.1	8.21192233	8.203710408	−0.1	69.21908301	−692.1908301	692.1908301
	0.1	8.21192233	8.220134252	0.1	69.21193364	692.1193364	692.1193364
	1	8.21192233	8.294041553	1	1268.190933	1268.190933	1268.190933
	5	8.21192233	8.622518447	5	6712.781412	1342.556282	1342.556282
I_0	−5	0.05752541	0.05464914	−5	1736.381837	−347.2763674	347.2763674
	−1	0.05752541	0.056950156	−1	275.4996097	−275.4996097	275.4996097
	−0.1	0.05752541	0.057467885	−0.1	6.312603238	−63.12603238	63.12603238
	0.1	0.05752541	0.057582935	0.1	6.304943567	63.04943567	63.04943567
	1	0.05752541	0.058100664	1	273.2583967	273.2583967	273.2583967
	5	0.05752541	0.060401681	5	1678.272644	335.6545288	335.6545288
n	−5	1.26338213	1.200213024	−5	39217.75222	−7843.550445	7843.550445
	−1	1.26338213	1.250748309	−1	6750.289073	−6750.289073	6750.289073
	−0.1	1.26338213	1.262118748	−0.1	570.1062165	−5701.062165	5701.062165
	0.1	1.26338213	1.264645512	0.1	565.2277079	5652.277079	5652.277079
	1	1.26338213	1.276015951	1	6257.216787	6257.216787	6257.216787
	5	1.26338213	1.326551237	5	27010.00522	5402.001044	5402.001044
R_S	−5	0.23712363	0.225267449	−5	713.3635006	−142.6727001	142.6727001
	−1	0.23712363	0.234752394	−1	89.36563136	−89.36563136	89.36563136
	−0.1	0.23712363	0.236886506	−0.1	1.282773334	−12.82773334	12.82773334
	0.1	0.23712363	0.237360754	0.1	1.281565178	12.81565178	12.81565178
	1	0.23712363	0.239494866	1	89.09322033	89.09322033	89.09322033
	5	0.23712363	0.248979812	5	705.5094879	141.1018976	141.1018976
R_P	−5	373.4745146	354.8007888	−5	15.63145821	−3.126291643	3.126291643
	−1	373.4745146	369.7397694	−1	0.618680164	−0.618680164	0.618680164
	−0.1	373.4745146	373.10104	−0.1	0.006077295	−0.060772951	0.060772951
	0.1	373.4745146	373.8479891	0.1	0.006090964	0.06090964	0.06090964
	1	373.4745146	377.2092597	1	0.594861151	0.594861151	0.594861151
	5	373.4745146	392.1482403	5	12.95791757	2.591583515	2.591583515

Appendix F. Contains the tables corresponding to the joint decimal truncation of two parameters

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Table F1
RMSE values for joint decimal truncation of two parameters for the RTC France solar cell (The values for RMSE have been multiplied by 10,000).

Parameter orparameters	Decimal places					
	8	7	6	5	4	3
I_{PV}	7.7300626905	7.7300627139	7.7300675148	7.7303896856	7.7698189484	10.4497981821
I_0	7.7300626905	7.7300626958	7.7300626958	7.7301138792	7.7521752817	9.0808222069
I_{PV} and I_0	7.7300626905	7.7300627105	7.7300673825	7.7303256876	7.7655035932	10.1013075994
I_{PV}	7.7300626905	7.7300627139	7.7300675148	7.7303896856	7.7698189484	10.4497981821
n	7.7300626905	7.7300626932	7.7300735978	7.7317618353	7.8630964480	9.6064228396
I_{PV} and n	7.7300626905	7.7300627230	7.7300846521	7.7327330014	7.9638321953	13.2816980689
I_{PV}	7.7300626905	7.7300627139	7.7300675148	7.7303896856	7.7698189484	10.4497981821
R_S	7.7300626905	7.7300627186	7.7300729761	7.7306135923	7.75517499998	10.6193759006
I_{PV} and R_S	7.7300626905	7.7300627239	7.7300723057	7.7306072896	7.7700939163	11.0506348169
I_0	7.7300626905	7.7300626958	7.7300626958	7.7301138792	7.7521752817	9.0808222069
n	7.7300626905	7.7300626932	7.7300735978	7.7317618353	7.8630964480	9.6064228396
I_0 and n	7.7300626905	7.7300626908	7.7300731353	7.7312236584	7.7772644308	7.7880688627
I_0	7.7300626905	7.7300626958	7.7300626958	7.7301138792	7.7521752817	9.0808222069
R_S	7.7300626905	7.7300627186	7.7300729761	7.7306135923	7.75517499998	10.6193759006
I_0 and R_S	7.7300626905	7.7300627291	7.7300730816	7.7307388411	7.7875632635	12.2787386488
n	7.7300626905	7.7300626932	7.7300735978	7.7317618353	7.8630964480	9.6064228396
R_S	7.7300626905	7.7300627186	7.7300729761	7.7306135923	7.75517499998	10.6193759006
n and R_S	7.7300626905	7.7300627178	7.7300798424	7.7319441638	7.8662024289	11.4063038416

Table F2
RMSE values for joint decimal truncation of two parameters for the MSX-60 solar module (The values for RMSE have been multiplied by 100).

Parameter orparameters	Decimal places				
	8	7	6	5	4
I_{PV}	1.1651382745	1.1651382835	1.1651383872	1.1651387681	1.1651630284
I_0	1.1651382745	1.1651382542	1.1651373475	1.1651362654	1.1672309129
I_{PV} and I_0	1.1651382745	1.1651382632	1.1651374364	1.1651360934	1.1669969008
I_{PV}	1.1651382745	1.1651382835	1.1651383872	1.1651387681	1.1651630284
n	1.1651382745	1.1651384963	1.1651412416	1.1652040799	1.1745626932
I_{PV} and n	1.1651382745	1.1651385057	1.1651414084	1.1652065060	1.1750965485
I_{PV}	1.1651382745	1.1651382835	1.1651383872	1.1651387681	1.1651630284
R_S	1.1651382745	1.1651382643	1.1651382440	1.1651374540	1.1651621662
I_{PV} and R_S	1.1651382745	1.1651382732	1.1651383555	1.1651377846	1.1651485897
I_0	1.1651382745	1.1651382542	1.1651373475	1.1651362654	1.1672309129
n	1.1651382745	1.1651384963	1.1651412416	1.1652040799	1.1745626932
I_0 and n	1.1651382745	1.1651384752	1.1651398145	1.1651718044	1.1676252536
I_0	1.1651382745	1.1651382542	1.1651373475	1.1651362654	1.1672309129
R_S	1.1651382745	1.1651382643	1.1651382440	1.1651374540	1.1651621662
I_0 and R_S	1.1651382745	1.1651382441	1.1651373244	1.1651371793	1.1677136527
n	1.1651382745	1.1651384963	1.1651412416	1.1652040799	1.1745626932
R_S	1.1651382745	1.1651382643	1.1651382440	1.1651374540	1.1651621662
n and R_S	1.1651382745	1.1651384856	1.1651411945	1.1651982302	1.1736844687

Table F3
RMSE values for joint decimal truncation of two parameters for the Photowatt-PWP 201 solar module (The values for RMSE have been multiplied by 1000).

Parameter orparameters	Decimal places					
	8	7	6	5	4	3
I_{PV}	2.0399922732	2.0399922746	2.0399923322	2.0400020358	2.0405541973	2.0615463210
I_0	2.0399922732	2.0399922732	2.0399922749	2.0399923844	2.0400235663	2.0412044545
I_{PV} and I_0	2.0399922732	2.0399922745	2.0399923249	2.0400011450	2.0404570215	2.0578266210
I_{PV}	2.0399922732	2.0399922746	2.0399923322	2.0400020358	2.0405541973	2.0615463210
n	2.0399922732	2.0399922747	2.0399928309	2.0400775114	2.0413995083	2.7151907635
I_{PV} and n	2.0399922732	2.0399922776	2.0399930606	2.0401143099	2.0427935470	2.8212109031
I_{PV}	2.0399922732	2.0399922746	2.0399923322	2.0400020358	2.0405541973	2.0615463210
R_S	2.0399922732	2.0399922732	2.0399922733	2.0399923428	2.0399951799	2.0403810717
I_{PV} and R_S	2.0399922732	2.0399922745	2.0399923299	2.0400013881	2.0405219309	2.0594315664
I_0	2.0399922732	2.0399922732	2.0399922749	2.0399923844	2.0400235663	2.0412044545
n	2.0399922732	2.0399922747	2.0399928309	2.0400775114	2.0413995083	2.7151907635
I_0 and n	2.0399922732	2.0399922745	2.0399927748	2.0400715164	2.0410116885	2.6692596402
I_0	2.0399922732	2.0399922732	2.0399922749	2.0399923844	2.0400235663	2.0412044545
R_S	2.0399922732	2.0399922732	2.0399922733	2.0399923428	2.0399951799	2.0403810717
I_0 and R_S	2.0399922732	2.0399922732	2.0399922752	2.0399924946	2.0400309044	2.0419121746
n	2.0399922732	2.0399922747	2.0399928309	2.0400775114	2.0413995083	2.7151907635
R_S	2.0399922732	2.0399922732	2.0399922733	2.0399923428	2.0399951799	2.0403810717
n and R_S	2.0399922732	2.0399922747	2.0399928276	2.0400766149	2.0413770877	2.7104727892

Table F4
RMSE values for joint decimal truncation of two parameters for the CdTe75669 solar module (The values for RMSE have been multiplied by 10000).

Parameter orparameters	Decimal places					
	8	7	6	5	4	3
I_{PV}	9.5068649100	9.5068649134	9.5068686402	9.5068809887	9.5185067190	10.0276311769
I_0	9.5068649100	9.5068649100	9.5068649098	9.5068649155	9.5068764500	9.5082049811
I_{PV} and I_0	9.5068649100	9.5068649134	9.5068686009	9.5068807085	9.5182349947	10.0093367878
I_{PV}	9.5068649100	9.5068649134	9.5068686402	9.5068809887	9.5185067190	10.0276311769
n	9.5068649100	9.5068649229	9.5068667009	9.5074109419	9.5660647033	12.7808701630
I_{PV} and n	9.5068649100	9.5068649292	9.5068722700	9.5074948148	9.5966041985	13.9042360599
I_{PV}	9.5068649100	9.5068649134	9.5068686402	9.5068809887	9.5185067190	10.0276311769
R_S	9.5068649100	9.5068649100	9.5068649099	9.5068649202	9.5068699134	9.5068928570
I_{PV} and R_S	9.5068649100	9.5068649132	9.5068685865	9.5068805056	9.5182580666	10.0238179197
I_0	9.5068649100	9.5068649100	9.5068649098	9.5068649155	9.5068764500	9.5082049811
n	9.5068649100	9.5068649229	9.5068667009	9.5074109419	9.5660647033	12.7808701630
I_0 and n	9.5068649100	9.5068649229	9.5068666306	9.5074066099	9.5644241648	12.6746611407
I_0	9.5068649100	9.5068649100	9.5068649098	9.5068649155	9.5068764500	9.5082049811
R_S	9.5068649100	9.5068649100	9.5068649099	9.5068649202	9.5068699134	9.5068928570
I_0 and R_S	9.5068649100	9.5068649100	9.5068649108	9.5068649430	9.5068935849	9.5085391927
n	9.5068649100	9.5068649229	9.5068667009	9.5074109419	9.5660647033	12.7808701630

(*continued on next page*)

Table F4 (*continued*)

Parameter orparameters	Decimal places					
	8	7	6	5	4	3
R_S	9.5068649100	9.5068649100	9.5068649099	9.5068649202	9.5068699134	9.5068928570
n and R_S	9.5068649100	9.5068649224	9.5068666469	9.5074067554	9.5652411234	12.7691799654

Table F5
RMSE values for joint decimal truncation of two parameters for the KC200GT solar module (The values for RMSE have been multiplied by 1000).

Parameter orparameters	Decimal places					
	8	7	6	5	4	3
I_{PV}	5.1031656936	5.1031656935	5.1031656989	5.1031660594	5.1032006868	5.1627897530
I_0	5.1031656936	5.1031657026	5.1031825184	5.1061017202	5.1675729566	17.6203742432
I_{PV} and I_0	5.1031656936	5.1031657016	5.1031821535	5.1060676158	5.1660739047	17.2481722836
I_{PV}	5.1031656936	5.1031656935	5.1031656989	5.1031660594	5.1032006868	5.1627897530
n	5.1031656936	5.1031657598	5.1031668860	5.1034819897	5.5534413560	11.4061942181
I_{PV} and n	5.1031656936	5.1031657616	5.1031669866	5.1034933885	5.5572191004	11.7775979220
I_{PV}	5.1031656936	5.1031656935	5.1031656989	5.1031660594	5.1032006868	5.1627897530
R_S	5.1031656936	5.1031656970	5.1031662070	5.1031814051	5.1038214235	5.1210459454
I_{PV} and R_S	5.1031656936	5.1031656965	5.1031661387	5.1031787805	5.1036698651	5.1405967102
I_0	5.1031656936	5.1031657026	5.1031825184	5.1061017202	5.1675729566	17.6203742432
n	5.1031656936	5.1031657598	5.1031668860	5.1034819897	5.5534413560	11.4061942181
I_0 and n	5.1031656936	5.1031657616	5.1031669866	5.1034933885	5.5572191004	11.7775979220
I_0	5.1031656936	5.1031657026	5.1031825184	5.1061017202	5.1675729566	17.6203742432
R_S	5.1031656936	5.1031656970	5.1031662070	5.1031814051	5.1038214235	5.1210459454
I_0 and R_S	5.1031656936	5.1031657110	5.1031873711	5.1064471665	5.1781756435	17.9416245671
n	5.1031656936	5.1031657598	5.1031668860	5.1034819897	5.5534413560	11.4061942181
R_S	5.1031656936	5.1031656970	5.1031662070	5.1031814051	5.1038214235	5.1210459454
n and R_S	5.1031656936	5.1031657508	5.1031662779	5.1033918375	5.5295878906	11.1220398742

Appendix G. Contains the tables corresponding to the simultaneous reduction of two parameters

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Table G1
Pairwise interaction effects of simultaneous parameter reduction for the RTC France solar cell using the WaOA parameter set.

Parameter pair	Reduction [%]	E_i [%]	E_j [%]	E_{ij} [%]	I_{ij} [%]	Interaction type
I_{PV}-I_0	−5	4296.29254	1324.384094	3877.389009	−1743.287625	compensating
	−2	1659.98068	471.8599733	1492.229545	−639.6111084	compensating
	−1	784.0885555	197.9407741	700.5861523	−281.4431772	compensating
I_{PV}-n	−5	4296.29254	23785.54245	25735.62463	−2346.210353	compensating
	−2	1659.98068	8597.153373	9448.576135	−808.5579184	compensating
	−1	784.0885555	4109.612236	4547.299955	−346.4008364	compensating
I_{PV}-R_S	−5	4296.29254	231.9975124	4201.002122	−327.287931	compensating
	−2	1659.98068	60.85015518	1617.526801	−103.304035	compensating
	−1	784.0885555	18.13089562	762.2395215	−39.97992959	compensating
I_{PV}-R_P	−5	4296.29254	8.706552046	4330.299382	25.30029016	cumulative
	−2	1659.98068	1.356918602	1673.152076	11.814477	cumulative
	−1	784.0885555	0.334135255	790.577058	6.154367323	cumulative
I_0-n	−5	1324.384094	23785.54245	22002.24105	−3107.68549	compensating
	−2	471.8599733	8597.153373	7976.143376	−1092.869971	compensating
	−1	197.9407741	4109.612236	3814.645431	−492.9075791	compensating
I_0-R_S	−5	1324.384094	231.9975124	1443.74637	−112.6352364	compensating
	−2	471.8599733	60.85015518	514.5476697	−18.16245878	compensating
	−1	197.9407741	18.13089562	217.7132344	1.64156473	cumulative
I_0-R_P	−5	1324.384094	8.706552046	1296.381203	−36.70944285	compensating
	−2	471.8599733	1.356918602	461.2023167	−12.01457522	compensating
	−1	197.9407741	0.334135255	192.9068377	−5.368071595	compensating
n-R_S	−5	23785.54245	231.9975124	24264.85444	247.3144771	cumulative
	−2	8597.153373	60.85015518	8640.72266	−17.28086839	compensating
	−1	4109.612236	18.13089562	4113.304703	−14.43842821	compensating
n-R_P	−5	23785.54245	8.706552046	23808.80021	14.55120734	cumulative
	−2	8597.153373	1.356918602	8607.154155	8.643862995	cumulative
	−1	4109.612236	0.334135255	4114.72412	4.777749231	cumulative
R_S-R_P	−5	231.9975124	8.706552046	212.6699474	−28.03411706	compensating
	−2	60.85015518	1.356918602	54.71170748	−7.495366301	compensating
	−1	18.13089562	0.334135255	16.08285669	−2.382174179	compensating

Table G2
Pairwise interaction effects of simultaneous parameter reduction for the MSX-60 solar module using the AWaOA parameter set.

Parameter pair	Reduction [%]	E_i [%]	E_j [%]	E_{ij} [%]	I_{ij} [%]	Interaction type
I_{PV}-I_0	−5	1275.454451	425.1530265	1001.824287	−698.7831907	compensating
	−2	457.2834174	127.5789776	349.1982207	−235.6641743	compensating
	−1	191.7633541	42.72826331	140.5942795	−93.89733784	compensating
I_{PV}-n	−5	1275.454451	10532.63183	11348.98455	−459.1017367	compensating
	−2	457.2834174	3645.604092	4007.163677	−95.72383225	compensating
	−1	191.7633541	1693.715671	1880.026294	−5.452730634	compensating
I_{PV}-R_S	−5	1275.454451	72.19372831	1181.033214	−166.6149657	compensating
	−2	457.2834174	14.54013944	417.6142486	−54.2093082	compensating
	−1	191.7633541	3.800816687	172.4949954	−23.06917542	compensating
I_{PV}-R_P	−5	1275.454451	0.165728098	1282.24918	6.629000638	cumulative
	−2	457.2834174	0.002300416	459.8743876	2.58866976	cumulative
	−1	191.7633541	−0.008679713	192.9841976	1.229523235	cumulative
I_0-n	−5	425.1530265	10532.63183	9817.262457	−1140.522404	compensating
	−2	127.5789776	3645.604092	3409.780048	−363.4030216	compensating
	−1	42.72826331	1693.715671	1584.137655	−152.3062788	compensating
I_0-R_S	−5	425.1530265	72.19372831	544.9809162	47.63416141	cumulative
	−2	127.5789776	14.54013944	171.0823476	28.96323054	cumulative
	−1	42.72826331	3.800816687	60.405653	13.876573	cumulative
I_0-R_P	−5	425.1530265	0.165728098	419.4626658	−5.856088723	compensating
	−2	127.5789776	0.002300416	125.5607953	−2.020482728	compensating
	−1	42.72826331	−0.008679713	41.93014727	−0.789436324	compensating
n-R_S	−5	10532.63183	72.19372831	10597.39769	−7.427874948	compensating
	−2	3645.604092	14.54013944	3620.338709	−39.80552156	compensating
	−1	1693.715671	3.800816687	1675.900575	−21.61591227	compensating
n-R_P	−5	10532.63183	0.165728098	10537.34201	4.544449628	cumulative
	−2	3645.604092	0.002300416	3647.650055	2.043663436	cumulative
	−1	1693.715671	−0.008679713	1694.761985	1.054993871	cumulative
R_S-R_P	−5	72.19372831	0.165728098	66.9070561	−5.452400307	compensating
	−2	14.54013944	0.002300416	13.30993187	−1.232507985	compensating
	−1	3.800816687	−0.008679713	3.458166138	−0.333970836	compensating

Table G3
Pairwise interaction effects of simultaneous parameter reduction for Photowatt-PWP 201 solar module using the CGTO parameter set.

Parameter pair	Reduction [%]	E_i [%]	E_j [%]	E_{ij} [%]	I_{ij} [%]	Interaction type
I_{PV}-I_0	−5	2009.692277	642.070993	1771.303824	−880.4594457	compensating
	−2	748.0168814	207.7038036	653.0268919	−302.6937932	compensating
	−1	332.6278614	76.13209736	286.1690897	−122.5908691	compensating
I_{PV}-n	−5	2009.692277	10243.60186	11256.01872	−997.2754146	compensating
	−2	748.0168814	3767.615762	4206.030664	−309.6019801	compensating
	−1	332.6278614	1790.96636	2015.66418	−107.9300411	compensating
I_{PV}-R_S	−5	2009.692277	144.8684047	1936.855288	−217.7053938	compensating
	−2	748.0168814	33.77466726	715.8281035	−65.96344522	compensating
	−1	332.6278614	9.388943846	316.3466486	−25.6701567	compensating
I_{PV}-R_P	−5	2009.692277	5.115167129	2038.433476	23.62603212	cumulative
	−2	748.0168814	0.785666594	759.0949444	10.29239631	cumulative
	−1	332.6278614	0.19303617	338.0009676	5.180069984	cumulative
I_0-n	−5	642.070993	10243.60186	9391.651981	−1494.020869	compensating
	−2	207.7038036	3767.615762	3457.737712	−517.5818537	compensating
	−1	76.13209736	1790.96636	1641.479173	−225.6192845	compensating
I_0-R_S	−5	642.070993	144.8684047	733.0684466	−53.87095111	compensating
	−2	207.7038036	33.77466726	240.0743565	−1.404114354	compensating
	−1	76.13209736	9.388943846	90.01898233	4.497941121	cumulative
I_0-R_P	−5	642.070993	5.115167129	620.1635368	−27.02262332	compensating
	−2	207.7038036	0.785666594	199.6553811	−8.834089139	compensating
	−1	76.13209736	0.19303617	72.6824848	−3.642649048	compensating
n-R_S	−5	10243.60186	144.8684047	10458.85368	70.38341328	cumulative
	−2	3767.615762	33.77466726	3785.053095	−16.33733449	compensating
	−1	1790.96636	9.388943846	1790.645394	−9.709909988	compensating
n-R_P	−5	10243.60186	5.115167129	10262.0011	13.28407438	cumulative
	−2	3767.615762	0.785666594	3775.500913	7.09948408	cumulative
	−1	1790.96636	0.19303617	1794.991831	3.832435237	cumulative
R_S-R_P	−5	144.8684047	5.115167129	130.198745	−19.78482684	compensating
	−2	33.77466726	0.785666594	29.66971925	−4.890614597	compensating
	−1	9.388943846	0.19303617	8.158736847	−1.423243169	compensating

Table G4
Pairwise interaction effects of simultaneous parameter reduction for the CdTe75669 solar module using the ERWCA parameter set.

Parameterpair	Reduction [%]	E_i [%]	E_j [%]	E_{ij} [%]	I_{ij} [%]	Interaction type
I_{PV}-I_0	−5	5448.028152	865.4471163	5161.418519	−1152.056749	compensating
	−2	2119.472231	293.2080856	2003.974107	−408.7062099	compensating
	−1	1012.838708	114.2802126	955.1155555	−172.0033649	compensating
I_{PV}-n	−5	5448.028152	12237.07514	14957.58804	−2727.515251	compensating
	−2	2119.472231	4496.762693	5652.926205	−963.3087192	compensating
	−1	1012.838708	2144.740052	2733.923784	−423.6549754	compensating
I_{PV}-R_S	−5	5448.028152	402.943618	5229.844438	−621.1273323	compensating
	−2	2119.472231	120.6880292	2027.100112	−213.0601478	compensating
	−1	1012.838708	40.22015718	965.9285444	−87.13032058	compensating
I_{PV}-R_P	−5	5448.028152	73.98963177	5576.147244	54.12946055	cumulative
	−2	2119.472231	14.22970445	2169.239682	35.53774635	cumulative
	−1	1012.838708	3.666962177	1037.419212	20.9135417	cumulative
I_0-n	−5	865.4471163	12237.07514	11113.1601	−1989.362152	compensating
	−2	293.2080856	4496.762693	4090.618724	−699.3520543	compensating
	−1	114.2802126	2144.740052	1949.069519	−309.9507454	compensating
I_0-R_S	−5	865.4471163	402.943618	1294.533674	26.14293973	cumulative
	−2	293.2080856	120.6880292	459.2648755	45.36876065	cumulative
	−1	114.2802126	40.22015718	191.7884878	37.28811802	cumulative
I_0-R_P	−5	865.4471163	73.98963177	787.1776932	−152.2590549	compensating
	−2	293.2080856	14.22970445	263.8598993	−43.57789078	compensating
	−1	114.2802126	3.666962177	101.0958335	−16.85134127	compensating
n-R_S	−5	12237.07514	402.943618	12063.34737	−576.6713814	compensating
	−2	4496.762693	120.6880292	4373.490727	−243.959995	compensating
	−1	2144.740052	40.22015718	2076.623889	−108.3363194	compensating
n-R_P	−5	12237.07514	73.98963177	12309.26295	−1.801815887	compensating
	−2	4496.762693	14.22970445	4527.060601	16.06820346	cumulative
	−1	2144.740052	3.666962177	2160.107928	11.70091468	cumulative
R_S-R_P	−5	402.943618	73.98963177	309.4882568	−167.4449929	compensating
	−2	120.6880292	14.22970445	88.40276605	−46.51496758	compensating
	−1	40.22015718	3.666962177	28.04287564	−15.84424372	compensating

Table G5
Pairwise interaction effects of simultaneous parameter reduction for the KC200GT solar module using the ERWCA parameter set.

Parameterpair	Reduction [%]	E_i [%]	E_j [%]	E_{ij} [%]	I_{ij} [%]	Interaction type
I_{PV}-I_0	−5	6739.229917	1736.381837	6017.779215	−2457.832539	compensating
	−2	2634.063125	633.091714	2344.069757	−923.0850819	compensating
	−1	1269.244542	275.4996097	1124.309819	−420.4343328	compensating
I_{PV}-n	−5	6739.229917	39217.75222	42499.25892	−3457.723217	compensating
	−2	2634.063125	14101.30949	15560.62026	−1174.752356	compensating
	−1	1269.244542	6750.289073	7504.884106	−514.6495084	compensating
I_{PV}-R_S	−5	6739.229917	713.3635006	6319.234418	−1133.358999	compensating
	−2	2634.063125	237.1044754	2455.995225	−415.1723757	compensating
	−1	1269.244542	89.36563136	1178.700547	−179.9096265	compensating
I_{PV}-R_P	−5	6739.229917	15.63145821	6793.69044	38.8290649	cumulative
	−2	2634.063125	2.50238457	2655.143935	18.5784255	cumulative
	−1	1269.244542	0.618680164	1279.653612	9.790390206	cumulative
I_0-n	−5	1736.381837	39217.75222	36852.61265	−4101.521411	compensating
	−2	633.091714	14101.30949	13289.06353	−1445.337674	compensating
	−1	275.4996097	6750.289073	6366.936965	−658.8517177	compensating
I_0-R_S	−5	1736.381837	713.3635006	2431.546119	−18.19921874	compensating
	−2	633.091714	237.1044754	904.5570691	34.3608797	cumulative
	−1	275.4996097	89.36563136	407.7090916	42.84385051	cumulative
I_0-R_P	−5	1736.381837	15.63145821	1698.959663	−53.0536322	compensating
	−2	633.091714	2.50238457	618.771508	−16.82259052	compensating
	−1	275.4996097	0.618680164	268.6185396	−7.499750267	compensating
n-R_S	−5	39217.75222	713.3635006	39350.30854	−580.8071798	compensating
	−2	14101.30949	237.1044754	13942.85724	−395.5567259	compensating
	−1	6750.289073	89.36563136	6647.643275	−192.0114289	compensating
n-R_P	−5	39217.75222	15.63145821	39247.97882	14.59513717	cumulative
	−2	14101.30949	2.50238457	14114.56474	10.75286788	cumulative
	−1	6750.289073	0.618680164	6757.107244	6.199490809	cumulative
R_S-R_P	−5	713.3635006	15.63145821	670.3067773	−58.68818151	compensating
	−2	237.1044754	2.50238457	221.1896794	−18.41718057	compensating
	−1	89.36563136	0.618680164	82.41392139	−7.570390135	compensating

Data availability

Data will be made available on request.

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